



BAHIR DAR UNIVERSITY
COLLEGE OF AGRICULTURE AND ENVIRONMENTAL SCIENCES
GRADUATE PROGRAM

**ADOPTION OF SUSTAINABLE LAND MANAGEMENT PRACTICES
AND ITS EFFECT ON HOUSEHOLD INCOME FROM THE EVIDENCE
FROM WEST BELESA DISTRICT, AMHARA REGION, ETHIOPIA**

M.SC. THESIS

BY: MULU ALI

JULY, 2026

BAHIR DAR, ETHIOPIA



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**A THESIS SUBMITTED IN PARTIAL FULFILLMENT OF THE
REQUIREMENTS FOR THE DEGREE OF MASTER OF SCIENCE
(M.SC.) IN RURAL DEVELOPMENT MANAGEMENT**

ADVISOR: GIRMA (PH.D.)

JULY, 2026

BAHIR DAR, ETHIOPIA

THESIS APPROVAL SHEET

As member of the Board of Examiners of the Master of Sciences (M.Sc.) thesis open defense examination, we have read and evaluated this thesis prepared by **Mr. mulu ali** entitled ” **Adoption Of Sustainable Land Management Practices And Its Effect On Household Income From The Evidence From West Belesa District, Amhara Region, Ethiopia**”. We hereby certify that, the thesis is accepted for fulfilling the requirements for the award of the degree of Master of Sciences (M.Sc.) in **Rural Development Management**.

Board of Examiners

----- Name of External Examiner	----- Signature	----- Date
----- Name of Internal Examiner	----- Signature	----- Date
----- Name of Chairman	----- Signature	----- Date

DECLARATION

This is to certify that this thesis entitled “**Adoption Of Sustainable Land Management Practices And Its Effect On Household Income From The Evidence From West Belesa District, Amhara Region, Ethiopia**”.”, Submitted in partial fulfillment of the requirements for the award of the degree of Master of Science in “**Rural Development Management**” to the Graduate Program of College of Agriculture and Environmental Sciences, Bahir Dar University by **Mr. mulu ali (ID. No. BDU1415269)** is an authentic work carried out by him under our guidance. The matter embodied in this project work has not been submitted earlier for an award of any degree or diploma to the best of our knowledge and belief.

Name of the student

Mulu ali Signature ----- Date, -----

Name of the supervisor

1. Girma (Ph.D.) Signature ----- Date, -----
Advisor

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DEDICATION

I dedicate this thesis manuscript to my late father, ali bikese, for his kindness, devotion, affection and unwavering support during my earlier period. Let God keep his soul in heaven.

ABBREVIATION

ADLI	Agricultural Development Led Industrialization
BDU	Bahir Dar University
CSA	Central Statistical Agency
DAs	Development Agent
FAO	Food and Agriculture Organization
GDP	Gross Domestic Product
HH	Household
IPCC	Intergovernmental Panel on Climate Change
MOA	Ministry of Agriculture
NGO	Non-Governmental Organization
PSNP	Productive Safety Net Program
SLM	Sustainable Land Management
SLMP	Sustainable Land Management Program
SPSS	Statistical Package for Social Sciences
TLU	Tropical Livestock Unit
WB	World Bank

TABLE OF CONTENTS

THESIS APPROVAL SHEET	I
DECLARATION.....	II
ACKNOWLEDGEMENT	III
DEDICATION	IV
ABBREVIATION	V
TABLE OF CONTENTS	VI
LIST OF TABLES	IX
LIST OF FIGURES	X
ABSTRACT.....	1
CHAPTER ONE	2
1. INTRODUCTION	2
1.1 Background of the Study	2
1.2 Statement of the Problem.....	3
1.3 Research objectives.....	4
1.3.1 General objective	4
1.3.2 Specific objectives	4
1.4 Research Questions.....	5
1.5 Significance of the Study	5
1.6 Scope of the Study	6
CHAPTER TWO	7
2. REVIEW OF RELATED LITERATURE	7
2.1 Introduction.....	7
2.2 Empirical Review.....	7
2.2.1 Empirical Studies on SLM Practices	7
2.2.2 Socio-Economic Benefits of SLM Practices.....	8
2.2.3 Factors Influencing the Adoption of SLM	9
2.2.4 Synthesis and Research Gaps.....	9
2.3 Definitions of Key Terms	10
2.4 Conceptual Framework.....	11
2.4.1 Key Components of the Framework.....	11
2.4.2 Interaction of Variables	13
CHAPTER THREE	14

3. RESEARCH METHODOLOGY.....	14
3.1 Introduction.....	14
3.2 Design of the Study.....	14
3.3 Description of Study Area	15
3.3.1 Geographical Location.....	15
3.3.2 Socio-Economic Profile	15
3.3.3 Environmental Characteristics	15
3.3.4 Map of the Study Area	16
3.3.5 Relevance to the Study.....	16
3.4 Participants of the Study	17
3.4.1 Target Population	17
3.4.2 Sample Size Determination.....	18
3.4.3 Sampling Technique.....	18
3.5 Data Sources	19
3.5.1 Primary Survey	19
3.5.2 Secondary Data Sources	20
3.6 Data Collection Tools/Methods	20
3.6.1 Structured Surveys	20
3.6.2 Stakeholder Interviews.....	20
3.7 Data Analysis.....	21
3.7.1 Quantitative Data Analysis	21
3.7.2 Qualitative Data Analysis	21
3.8 Variables	21
3.8.1 Independent Variables.....	21
3.8.2 Dependent Variable.....	22
3.9 Model Specification (Logit).....	22
3.10 Validity and Reliability Test / Data Trustworthiness	22
3.11 Research Protocol with Human Participants.....	23
CHAPTER FOUR.....	24
RESULTS AND DISCUSSION	24
4.1 Introduction.....	24
4.2 Socio-Economic and Demographic Characteristics of Households	24
4.2.1 Continuous Household Characteristics	24
4.2.2 Categorical Household Characteristics	26

4.3 Status of Sustainable Land Management (SLM) Practices.....	28
4.3.2 Categorization of SLM Adoption Status.....	30
4.3.3 Adoption Patterns by Kebele Status	31
4.4 Determinants of Sustainable Land Management Adoption	32
4.5 Effect of SLM Practices on Household Income and Food Security	35
4.5.1 Agricultural Productivity Trends and Inputs	36
4.5.2 Household Food Security.....	37
4.6 Qualitative Data Analysis	37
4.6.1 Summary of Qualitative Findings.....	42
CHAPTER FIVE	44
SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS.....	44
5.1 Summary	44
5.2 Conclusions.....	45
5.3 Recommendations.....	46
REFERENCES	48
APPENDIX.....	51
General information	51

LIST OF TABLES

Table 1 sample size distribution	19
Table 2: Comparison of Continuous Household Characteristics by SLM Adoption Status....	24
Table 3: Categorical Household Characteristics by SLM Adoption Status.....	26
Table 4: Adoption of SLM Practices by Adoption Status	28
Table 5: Distribution of Households by SLM Adoption Status.....	30
Table 6: SLM Adoption Distribution by Kebele Status.....	31
Table 7: Factors Affecting SLM Adoption - Logistic Regression Results	32
Table 8: Comparison of Annual Household Income by SLM Adoption Status	35
Table 9: Income Change After SLM Adoption by Adoption Status	35
Table 10: Agricultural Productivity Trends by SLM Adoption Status.....	36
Table 11: Food Shortage by SLM Adoption Status.....	37
Table 12: Summary of Qualitative Themes and Key Findings.....	42

LIST OF FIGURES

Figure 1 conceptual framework	13
Figure 2 map of the study area.....	16

ABSTRACT

Sustainable Land Management (SLM) has gained global recognition as a vital strategy to address the twin challenges of land degradation and rural poverty. In Ethiopia, where over 85% of the population depends on agriculture, land degradation, deforestation, and climate variability pose significant threats to rural livelihoods and food security. Despite the implementation of various SLM interventions, empirical evidence on their adoption and socio-economic impacts at the household level remains limited, particularly in localized contexts. This study was conducted to assess the adoption of Sustainable Land Management (SLM) practices and their effect on household income in West Belesa District, Amhara Region, Ethiopia. The specific objectives were to: (1) identify the sustainable land management practices adopted by local communities in the study area; (2) analyze the effect of SLM practices on household income; and (3) examine the factors that facilitate or hinder the adoption of sustainable land management practices.

A cross-sectional survey design with a mixed-methods approach was employed. Primary data were collected from 193 randomly selected households using structured questionnaires, supplemented by key informant interviews and focus group discussions. The sample size was determined using Yamane's (1967) formula at 95% confidence level and 7% margin of error. A three-stage sampling technique was used: stratification of kebeles into treated and control groups, selection of four kebeles (Kalay, Asawogari, Talla, and Ebrareg) using simple random sampling, and proportional allocation of households within selected kebeles. Both quantitative and qualitative data were collected. Quantitative data were analyzed using descriptive statistics (frequencies, percentages, means, and standard deviations) and a binary logistic regression model with the aid of SPSS software. Qualitative data from key informant interviews were analyzed using thematic analysis. The study found that the majority of households adopted SLM practices, with crop rotation and terracing being the most widely implemented practices. Adopters had significantly higher household income, better food security, and greater access to extension services, credit, and training compared to non-adopters. Logistic regression analysis revealed that institutional factors such as extension access, credit availability, and training participation positively influenced adoption, along with household characteristics including education, farm size, and livestock ownership. Conversely, age and family size had negative effects on adoption. Qualitative findings further highlighted financial constraints and labor shortages as major barriers limiting adoption. The study confirms that

SLM practices contribute meaningfully to improved household welfare and agricultural productivity.

CHAPTER ONE

1. INTRODUCTION

1.1 Background of the Study

Sustainable management of natural resources has become a global priority in the face of increasing land degradation, climate change, and rising population pressure. Globally, about one-third of the world's land is degraded, threatening food security, biodiversity, and the livelihoods of millions of people (FAO, 2021; IPCC, 2022). Sustainable Land Management (SLM) practices such as agroforestry, soil and water conservation, and watershed management are widely promoted as essential strategies for achieving the Sustainable Development Goals (SDGs), particularly SDG 2 (Zero Hunger), SDG 13 (Climate Action), and SDG 15 (Life on Land) (UN, 2020). Despite global commitments, the pace of land degradation remains alarming, especially in developing countries where agriculture is the backbone of rural livelihoods.

In Sub-Saharan Africa, natural resource degradation has intensified food insecurity and rural poverty. Studies indicate that soil erosion, deforestation, and recurrent droughts significantly constrain agricultural productivity, which employs about 60% of the population in the region (World Bank, 2021; UNEP, 2022). While countries have adopted climate-smart agriculture and sustainable land management interventions, adoption levels remain uneven due to limited institutional capacity, resource constraints, and socio-economic barriers (FAO, 2022). As a result, millions of smallholder farmers face increasing vulnerability to climate shocks and declining agricultural yields.

Ethiopia is among the countries most affected by land degradation and climate variability. More than 85% of Ethiopians depend on agriculture, which contributes about one-third of GDP, yet the sector is highly vulnerable to environmental degradation and erratic rainfall (CSA, 2021; World Bank, 2022). In response, Ethiopia has implemented large-scale programs such as the Sustainable Land Management Project (SLMP) and Green Legacy Initiative, which emphasize watershed restoration, soil conservation, and tree planting (MoA, 2021; FAO, 2022). Recent evidence shows that these interventions have improved soil fertility, reduced erosion, and enhanced food security in some areas (Gebreselassie et al., 2022). However, challenges remain, including limited adoption at household level, weak extension systems, and uneven socio-economic benefits.

In the Amhara region, where agriculture is predominantly rain-fed, land degradation and deforestation are widespread. Studies report that about 27% of the region's land is highly degraded due to unsustainable farming practices, overgrazing, and population pressure (Amhara Bureau of Agriculture, 2021). Watershed management and agroforestry initiatives have been introduced, with some success in reducing soil erosion and enhancing crop yields (Gebremariam et al., 2022). Nevertheless, adoption rates remain low, particularly among resource-poor farmers, due to financial and labor constraints, and lack of awareness.

West Belesa District, the focus of this study, exemplifies these challenges. Located in a semi-arid zone of Central Gondar Zone, the district experiences recurrent drought, soil erosion, and low agricultural productivity. While various SLM interventions have been introduced—including terracing, tree planting, and community-based watershed management—their socio-economic impacts remain underexplored. Farmers continue to face persistent food insecurity, declining incomes, and vulnerability to climate shocks (MoA, 2022; CSA, 2023). Understanding which practices are adopted, their socio-economic benefits, and the barriers to adoption is therefore crucial for designing effective, context-specific policies and interventions that can strengthen resilience and improve livelihoods in West Belesa District.

1.2 Statement of the Problem

Sustainable Land Management (SLM) has gained global recognition as a vital strategy to address the twin challenges of land degradation and rural poverty. Globally, one-third of agricultural land is degraded, and threatening the livelihoods of millions (FAO, 2021; IPCC, 2022). Studies show that SLM practices such as agroforestry, terracing, and watershed management can restore soil fertility, increase resilience, and improve rural incomes (UNEP, 2022). However, evidence also indicates that adoption and impact vary widely across regions, with limited clarity on the socio-economic outcomes at household level (World Bank, 2021).

In Sub-Saharan Africa, natural resource degradation remains one of the most pressing development challenges. Soil erosion, deforestation, and erratic rainfall have severely constrained agricultural productivity, which directly supports over 60% of the population (NEPAD, 2021). While several studies document the potential of SLM to improve yields and resilience, they also reveal persistent barriers to adoption, including insecure land tenure, labor shortages, and inadequate extension services (Mekuria et al., 2015; Doss et al., 2018). Moreover, there is still limited empirical evidence on how these practices translate into concrete socio-economic outcomes, such as income across diverse local contexts in Africa.

In Ethiopia, natural resource degradation is particularly severe due to rapid population growth, deforestation, and unsustainable farming practices. National initiatives such as the Sustainable

Land Management Project (SLMP) and the Green Legacy Initiative have promoted soil and water conservation, afforestation, and watershed restoration (MoA, 2021; FAO, 2022). Research confirms that these interventions have reduced soil erosion and improved crop yields (Gebreselassie et al., 2022). Nevertheless, the socio-economic impacts of SLM at the household level remain inconsistently documented. Most studies are conducted at regional or national levels, leaving gaps in understanding localized effects on rural livelihoods, particularly in terms of income (CSA, 2021).

The Amhara Region of Ethiopia is among the most affected by land degradation, where agriculture is mainly rain-fed and highly vulnerable to climate shocks. Although soil and water conservation measures and watershed management projects have been widely implemented, adoption remains uneven, and evidence of their socio-economic contributions is limited (Amhara National Regional State Bureau of Agriculture, 2021; Gebremariam et al., 2022). In West Belesa District, natural resource degradation, recurrent drought, and food insecurity are persistent challenges. While interventions such as terracing, afforestation, and watershed management have been introduced, no comprehensive study has yet examined the extent of adoption, the socio-economic impacts on households, and the barriers to wider uptake.

Therefore, the gap this study seeks to fill lies in providing context-specific, empirical evidence on the adoption of SLM practices and their dual socio-economic impacts on household income in West Belesa District. While income is a key determinant, of a multi-dimensional outcome also influenced by production, stability, and utilization. This study intentionally examines them concurrently to assess whether and how SLM practices contribute to improved economic welfare and nutritional well-being, and to explore the potential linkages between these two critical indicators in the local context. By focusing on this local context, the study contributes to bridging the broader literature gap between national-level assessments and localized realities, thereby generating integrated insights that are critical for designing holistic and effective policies and interventions.

1.3 Research objectives

1.3.1 General objective

The general objective of the study is to assess the adoption of sustainable land management practices and their effect on household income in West Belesa District.

1.3.2 Specific objectives

1. To identify the sustainable Land management practices adopted by local communities in the study area.
2. To analyze the effect of SLM practice on household income..

3. To examine the factors that facilitates or hinders the adoption of sustainable Land management practice.

1.4 Research Questions

- Which sustainable Land management practices are most commonly implemented by local communities in West Belesa District?
- What is the effect of SLM practices on household income?
- What factors facilitate or constrain the adoption of sustainable Land management practices in the district?

1.5 Significance of the Study

This study offers valuable insights into sustainable Land management (SLM) within the specific context of West Belesa District. By exploring how local communities adopt and utilize SLM practices, as well as the socio-economic outcomes of these interventions, the research fills an important gap in the literature regarding context-specific strategies for environmental conservation and rural development. It thereby contributes to the academic body of knowledge by providing evidence-based findings that can guide further studies on best practices, sustainability challenges, and livelihood impacts in similar agro-ecological regions.

In practical terms, the research finding was beneficial to multiple stakeholders. Local government authorities was gain a clearer understanding of the community-level barriers and facilitators of SLM, enabling them to design more targeted policies, training programs, and resource allocations that align with local realities. Non-governmental organizations (NGOs) and development agencies can utilize the study's recommendations to refine their interventions and enhance project effectiveness, ensuring that efforts to conserve natural resources are integrated with measures that improve food security and household income. Community members, including smallholder farmers and local youth, was indirectly benefit from improved policy-making and programmatic support, as these can lead to more sustainable livelihoods, increased resilience to climate shocks, and better overall welfare.

Ultimately, the dissertation addresses both the academic and applied dimensions of sustainable resource management. By highlighting successful SLM practices and identifying actionable strategies to enhance local governance and community participation, the study's outcomes can serve as a practical roadmap for scaling up sustainable interventions. Its localized focus ensures that recommendations are tailored to the socio-economic and ecological conditions of West Belesa District, while also offering broader lessons that can be adapted to other districts and regions with similar resource management challenges.

1.6 Scope of the Study

The scope of this study is delineated by several parameters, primarily geographical, thematic, and methodological. Geographically, the research focuses on West Belesa District in the Amhara Region, Ethiopia. This district has been selected due to its diverse agro-ecological zones, existing resource management interventions, and pressing challenges related to soil degradation and water scarcity. The findings from this specific locale was offer insights into the unique socio-economic and environmental context of rural communities in the Ethiopian highlands.

From a thematic perspective, the study centers on sustainable Land management (SLM), exploring key practices such as soil and water conservation, agroforestry, and integrated watershed management. The research analyzes both the socio-economic impacts of these practices focusing on indicators like household income and the factors influencing their adoption (e.g., governance structures, policy frameworks, and local knowledge). While environmental and ecological dimensions (such as changes in vegetation cover or biodiversity) may be referenced, the primary emphasis remains on socio-economic outcomes and community-level dynamics.

Methodologically, the study adopts a mixed-methods approach incorporating both quantitative and qualitative data. Household surveys, key informant interviews, and focus group discussions was conducted to gather empirical evidence about the adoption, impact, and sustainability of SLM practices. The population of interest includes local households, community leaders, and relevant government and non-governmental stakeholders. However, detailed scientific measurements such as soil nutrient analysis or detailed biodiversity assessments fall outside the direct scope of this research.

Overall, the study's scope is intentionally focused to ensure depth and feasibility of investigation within the given time and resource constraints. By narrowing the research to West Belesa District and emphasizing the socio-economic dimensions of SLM, the study aims to produce contextually rich insights and actionable recommendations.

CHAPTER TWO

2. REVIEW OF RELATED LITERATURE

2.1 Introduction

The purpose of this chapter is to provide a comprehensive review of the existing literature on sustainable land management (SLM), with a particular focus on its socio-economic implications. By synthesizing theoretical perspectives, global experiences, and region-specific studies, this review aims to establish a solid conceptual foundation and contextual backdrop for the subsequent empirical investigation. In doing so, it seeks to illustrate both the evolution of SLM as a field of study and the current debates that surround its practice, particularly in agrarian communities such as those found in West Belesa District.

The discussion begins by examining key definitions and conceptual frameworks that have shaped the discourse on SLM, highlighting how these frameworks have shifted from a primarily conservation-oriented focus to a more holistic, people-centered approach (FAO, 2019; UNEP, 2019). The chapter then explores the socio-economic dimensions of resource management, drawing on evidence from global, African, and Ethiopian contexts to identify both successful interventions and persisting challenges. Through this integrative review, gaps and inconsistencies in the existing body of knowledge are revealed, thereby underscoring the need for localized, empirical research that examines how SLM practices can effectively improve livelihoods and foster sustainable development at the grassroots level

2.2 Empirical Review

In line with the specific objectives of this study, the empirical review is structured into three subsections: (i) studies on SLM practices, (ii) socio-economic benefits of these practices, and (iii) factors influencing adoption. This approach ensures that the literature review directly addresses the research questions and clarifies the knowledge gaps that the present study seeks to fill.

2.2.1 Empirical Studies on SLM Practices

Globally, a wide range of SLM practices have been documented in both developed and developing countries. Techniques such as integrated watershed management, agroforestry, conservation tillage, and soil and water conservation have proven effective in mitigating land degradation and enhancing productivity (UNEP, 2019). For example, India's Integrated Watershed Management Programme has demonstrated improvements in soil fertility and water retention, leading to higher agricultural yields (UNEP, 2019). In Latin America, agroforestry

systems are commonly adopted to restore degraded landscapes and improve biodiversity, showcasing the global adaptability of such practices.

In Africa, community-based natural resource management (CBNRM) has emerged as a cornerstone of sustainable resource use. In Namibia and Kenya, CBNRM has empowered local communities to manage forests and grazing lands, yielding both ecological restoration and livelihood benefits (Jones & Murphree, 2004). Similarly, soil and water conservation practices, such as terracing and check dams, have been widely promoted in East Africa to combat erosion and land degradation (Mekuria et al., 2015). These studies illustrate the diversity of practices while also highlighting the unevenness of their adoption across regions.

Ethiopian experiences further demonstrate the widespread promotion of SLM practices through national programs such as the Sustainable Land Management Project (SLMP). Practices like terracing, reforestation, and community-based forest management have been shown to reduce soil erosion and improve agricultural productivity in targeted areas (Hurni et al., 2015; MoA, 2016). However, adoption levels vary across districts due to ecological, institutional, and socio-cultural factors. This underscores the importance of localized studies like the present one, which focuses on identifying the specific practices adopted in West Belesa District.

2.2.2 Socio-Economic Benefits of SLM Practices

Evidence from different parts of the world shows that SLM practices can generate significant socio-economic benefits. FAO (2019) emphasizes that sustainable land management improves agricultural productivity, thereby enhancing food security and rural incomes. Similarly, the World Bank (2020) reports that investments in SLM lead to greater resilience against climate shocks and long-term poverty reduction. These findings suggest that environmental and economic objectives can be mutually reinforcing when practices are effectively implemented. In Africa, CBNRM initiatives have produced tangible welfare improvements. For instance, in Namibia, community-managed wildlife conservation has boosted local incomes through eco-tourism while also restoring biodiversity (Jones & Murphree, 2004). In Ethiopia and Kenya, studies by Mekuria et al. (2015) indicate that soil and water conservation practices not only reduce erosion but also improve crop yields, thereby contributing to household food security. Yet, benefits are not always equally distributed, as poorer households often face barriers to fully participating in or benefiting from these interventions.

Specifically in Ethiopia, watershed management programs have been shown to improve water availability, reduce flooding, and enhance agricultural outputs (Gebregziabher et al., 2016). Agroforestry has also provided alternative income streams through timber, fuelwood, and fruit production. However, CSA (2019) notes that the socio-economic benefits are uneven across

regions, with some communities gaining more than others due to differences in market access, land quality, and institutional support. This variability highlights the need for district-level evidence to assess whether households in West Belesa are experiencing comparable benefits, particularly in terms of income and food security.

2.2.3 Factors Influencing the Adoption of SLM

The literature indicates that adoption of SLM practices is shaped by a range of socio-economic, institutional, and ecological factors. Globally, strong policy support, financial incentives, and secure land tenure are identified as key enablers of adoption (World Bank, 2020; FAO, 2020). Conversely, barriers such as high labor demands, upfront costs, and limited technical knowledge frequently constrain adoption, especially among smallholder farmers. These findings point to the importance of institutional support and localized capacity-building in ensuring sustainable uptake.

In Sub-Saharan Africa, adoption challenges are often linked to weak governance, gender inequalities, and lack of access to credit or extension services (Mekuria et al., 2015; Doss et al., 2018). Women, in particular, face structural barriers to participation in resource management, despite their critical role in managing water, fuel wood, and household food systems. At the same time, positive cases, such as Rwanda's climate-smart agriculture programs, show that when communities receive adequate training and institutional support, adoption rates increase significantly.

Ethiopian studies emphasize that land tenure security, extension services, and community participation play decisive roles in shaping adoption (Deininger et al., 2011; Bekele & Drake, 2003). However, challenges such as unclear property rights, inadequate financial support, and cultural resistance have hindered long-term sustainability of interventions (CSA, 2019). This suggests that beyond technical solutions, attention must be given to institutional reforms and socio-cultural dynamics. The present study therefore seeks to explore both enabling factors and barriers specific to West Belesa District, where adoption levels remain uneven.

2.2.4 Synthesis and Research Gaps

Overall, the empirical review highlights three main insights. First, while a range of SLM practices have been promoted globally and nationally, their actual adoption remains highly context-dependent. Second, empirical evidence points to significant socio-economic benefits of SLM practices, though these are not uniformly experienced across households and regions. Third, adoption is strongly shaped by both enabling and constraining factors, including land tenure, institutional capacity, and cultural dynamics.

Despite these findings, important gaps remain. Much of the literature in Ethiopia is based on large-scale program assessments, with limited attention to localized, district-level contexts. Moreover, few studies simultaneously examine practices, socio-economic outcomes, and adoption factors within a single analytical framework. This study addresses these gaps by focusing on West Belesa District, thereby contributing context-specific evidence that can inform both national policy and local interventions.

2.3 Definitions of Key Terms

A clear understanding of the key concepts used in this study is essential for situating the research within existing scholarship. The following definitions are derived from widely accepted literature and institutional reports to ensure scientific rigor and consistency with the broader review.

Sustainable Land Management (SLM): SLM refers to the stewardship and use of natural resources such as land, water, and forests in ways that meet present social, economic, and environmental needs without compromising the capacity of future generations to meet theirs. It encompasses practices like soil and water conservation, agroforestry, and community-based conservation (FAO, 2020; UNEP, 2019).

Socio-Economic Impact: Socio-economic impact denotes the measurable effects of policies or interventions on individuals and communities. These are typically assessed using indicators such as income levels, food security, education, health, and overall quality of life (World Bank, 2020; CSA, 2019).

Food Security: Food security exists when all people, at all times, have physical, social, and economic access to sufficient, safe, and nutritious food that meets their dietary needs and preferences for an active and healthy life (FAO, 2001; FAO, 2019).

Livelihood Diversification: This refers to the strategies by which households or individuals generate income from multiple sources to reduce vulnerability and enhance resilience, often combining farming with non-farm activities such as trade, handicrafts, or wage labor (Scoones, 1998; DFID, 1999).

Resilience (in Rural Context): Resilience is the capacity of individuals, households, or communities to anticipate, absorb, and recover from environmental and socio-economic shocks such as droughts, floods, or market fluctuations—while maintaining or improving their well-being (IPCC, 2019; Scoones, 2015).

Agroforestry: Agroforestry is a land-use management system that integrates trees and shrubs with crops and/or livestock on the same parcel of land, thereby enhancing productivity, biodiversity, and sustainability over time (UNEP, 2019; Hurni et al., 2015).

Watershed Management: Watershed management is an integrated approach to the conservation and utilization of soil, water, and vegetation resources within a watershed, with the aim of sustaining ecosystem services and improving productivity (MoA, 2016; Gebregziabher et al., 2016).

2.4 Conceptual Framework

The conceptual framework serves as a guiding structure for this study, illustrating the relationships between key variables associated with sustainable natural resource management (SLM) and their socio-economic impacts. The framework integrates insights from theoretical perspectives, empirical studies, and contextual factors to establish a systematic approach for analyzing the dynamics of SLM in West Belesa District. It emphasizes the interaction between management practices, facilitating factors, barriers, and outcomes, providing a basis for understanding how these elements collectively influence resource sustainability and community well-being.

2.4.1 Key Components of the Framework

The framework consists of four main components:

1. **SLM Practices:** These include specific interventions such as soil and water conservation, agroforestry, terracing, and watershed management. These practices represent the independent variables and are central to achieving resource sustainability and resilience.
2. **Facilitating Factors:** These factors enable the successful implementation of SLM practices. They include community participation, institutional support, technical knowledge, financial resources, and favorable policies.
3. **Barriers to Adoption:** These represent challenges that hinder the effective implementation and sustainability of SLM practices. Common barriers include land tenure insecurity, insufficient funding, limited capacity, and socio-cultural resistance.
4. **Socio-Economic Outcomes:** These are the dependent variables and reflect the impact of SLM on household income, food security, resource availability, and community well-being.

The conceptual framework hypothesizes that the adoption of SLM practices leads to improved socio-economic outcomes when supported by facilitating factors and minimal barriers. For example, watershed management practices can enhance water availability and agricultural productivity, but their success depends on community participation, access to technical knowledge, and government support. Conversely, barriers such as inadequate infrastructure or

unclear property rights may limit the effectiveness of these interventions, reducing their socio-economic benefits.

A critical aspect of the framework is the feedback loop between outcomes and adoption. Improved socio-economic outcomes, such as increased household income and food security, can incentivize broader adoption of SLM practices. However, if practices fail to deliver tangible benefits, they may be abandoned, highlighting the importance of achieving measurable impacts to sustain participation.

The framework recognizes that SLM practices and outcomes are shaped by contextual factors, including ecological conditions, cultural norms, and governance structures. In West Belesa District, factors such as soil quality, rainfall variability, and local governance capacity play a significant role in determining the feasibility and effectiveness of resource management strategies. For instance, regions with highly degraded land may require intensive rehabilitation efforts, while areas with strong traditional leadership may benefit from enhanced community mobilization.

This conceptual framework provides a structured approach for examining the relationships between SLM practices, barriers, facilitating factors, and socio-economic outcomes in West Belesa District. It guides the formulation of research questions, data collection, and analysis, ensuring that the study captures the complex dynamics of resource management. By focusing on these interactions, the framework helps identify actionable strategies for enhancing the adoption and impact of SLM practices, ultimately contributing to sustainable development in the region.

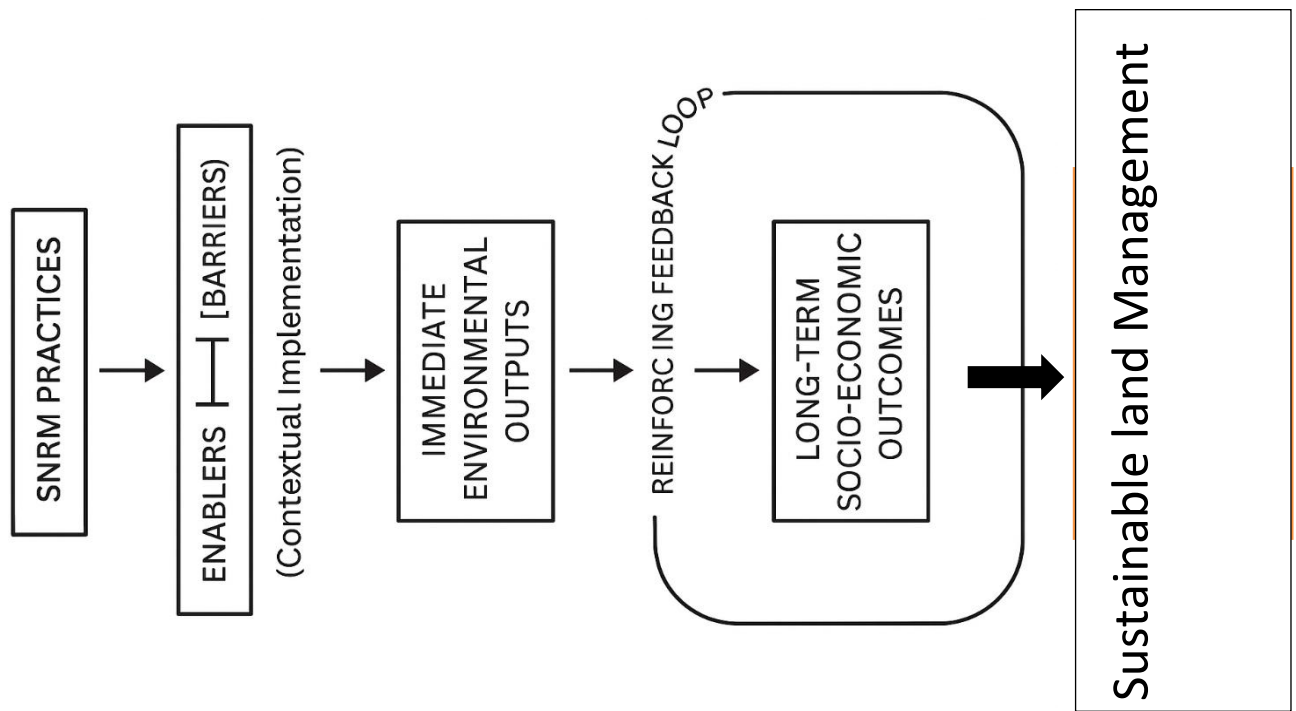


Figure 1 conceptual framework

Diagram of the Conceptual Framework

The framework can be visualized as a flow diagram with the following components:

1. Inputs: SLM practices (e.g., terracing, agroforestry).
2. Enablers and Barriers: Facilitating factors (e.g., policies, community participation) and barriers (e.g., financial constraints).
3. Processes: Implementation of practices influenced by contextual factors.
4. Outputs: Immediate impacts (e.g., reduced erosion, increased water retention).
5. Outcomes: Long-term socio-economic benefits (e.g., income, food security).

2.4.2 Interaction of Variables

In conclusion, the conceptual framework integrates multiple dimensions of SLM to provide a comprehensive understanding of its implementation and impacts in West Belesa District. It emphasizes the interconnectedness of practices, enabling factors, challenges, and outcomes, offering a holistic lens through which to analyze the study's objectives. By applying this framework, the research can generate nuanced insights that inform both academic inquiry and practical interventions for sustainable Land management.

CHAPTER THREE

3. RESEARCH METHODOLOGY

3.1 Introduction

This chapter outlines the research methodology employed to examine the practices and socio-economic impacts of sustainable Land management (SLM) in West Belesa District. It begins by presenting the research design, emphasizing the use of a descriptive and mixed-methods approach to capture both quantitative and qualitative data. The chapter also details the study area, population, sampling techniques, and data collection methods, including household surveys, key informant interviews, and focus group discussions. A data analysis technique for quantitative and qualitative data was discussed to ensure a comprehensive understanding of the findings. Ethical considerations are addressed to maintain the integrity of the research and protect participant rights. Overall, the methodology provides a systematic framework for achieving the study's objectives and generating reliable, actionable insights.

3.2 Design of the Study

The study employs a descriptive and mixed-methods design, integrating quantitative and qualitative approaches. This design is chosen to provide a comprehensive understanding of the practices and socio-economic impacts of sustainable Land management (SLM) in West Belesa District. The descriptive design is appropriate as it enables the systematic exploration and documentation of SLM practices, barriers, and facilitating factors within the local context.

A quantitative method was applied through structured household surveys, which measure key variables such as household income, food security, and participation in SLM practices. These methods ensure the collection of statistically robust data that can be analyzed to identify patterns, relationships, and trends. A qualitative method, on the other hand, was employed via key informant interviews and focus group discussions to capture in-depth insights into community perceptions, lived experiences, and socio-cultural dynamics influencing SLM.

The combination of quantitative and qualitative approaches is justified as it allows for a holistic understanding of the research problem. Quantitative data provides objective, measurable outcomes, while qualitative data adds contextual richness and depth. This integration ensures that both the "what" and "why" of SLM practices and their impacts were addressed.

Additionally, the study adopts a cross-sectional design, collecting data at a single point in time to provide a snapshot of SLM practices and their impacts. This design is cost-effective and practical for understanding the current state of resource management in the district. The

descriptive, mixed-methods, and cross-sectional design collectively enable the study to address its objectives comprehensively while ensuring reliability and relevance of the findings.

3.3 Description of Study Area

This section provides an overview of the study area, West Belesa District, located in the Amhara Region of Ethiopia. The description highlights the geographical, socio-economic, and environmental characteristics of the district, as these factors are critical to understanding the dynamics of sustainable Land management (SLM) within the local context.

3.3.1 Geographical Location

West Belesa District is located in the Central Gondar Zone of the Amhara Region, Ethiopia. It is situated approximately 368.5 kilometers southeast of Bahir Dar, the regional capital. The district encompasses diverse topography, including highlands and lowlands, contributing to various agro-ecological zones. The altitude ranges from approximately 1,743 meters above sea level, influencing the local climate and agricultural practices (African Development Bank Group, 2022).

3.3.2 Socio-Economic Profile

The population of West Belesa District is predominantly rural, with livelihoods primarily dependent on subsistence agriculture. Major crops grown include teff, maize, sorghum, and barley, while livestock rearing plays a supplementary role in household income. The district faces significant socio-economic challenges, including food insecurity, limited access to markets, and high rates of poverty. Infrastructure, such as roads, schools, and healthcare facilities, remains underdeveloped, further limiting opportunities for economic diversification and development. These socio-economic conditions shape the adoption and effectiveness of SLM practices in the district.

3.3.3 Environmental Characteristics

West Belesa District is highly susceptible to environmental degradation, including soil erosion, deforestation, and water scarcity. The semi-arid climate, combined with unsustainable land-use practices such as overgrazing and improper farming techniques, exacerbates resource depletion. Efforts to address these challenges include terracing, reforestation programs, and watershed management initiatives led by government and non-governmental organizations. Despite these efforts, the district continues to face significant barriers to sustainable resource use, such as limited technical capacity, land tenure insecurity, and low community participation.

3.3.4 Map of the Study Area

To provide a visual representation of the study area, below is a map highlighting the location of West Belesa District within the Amhara Region of Ethiopia.

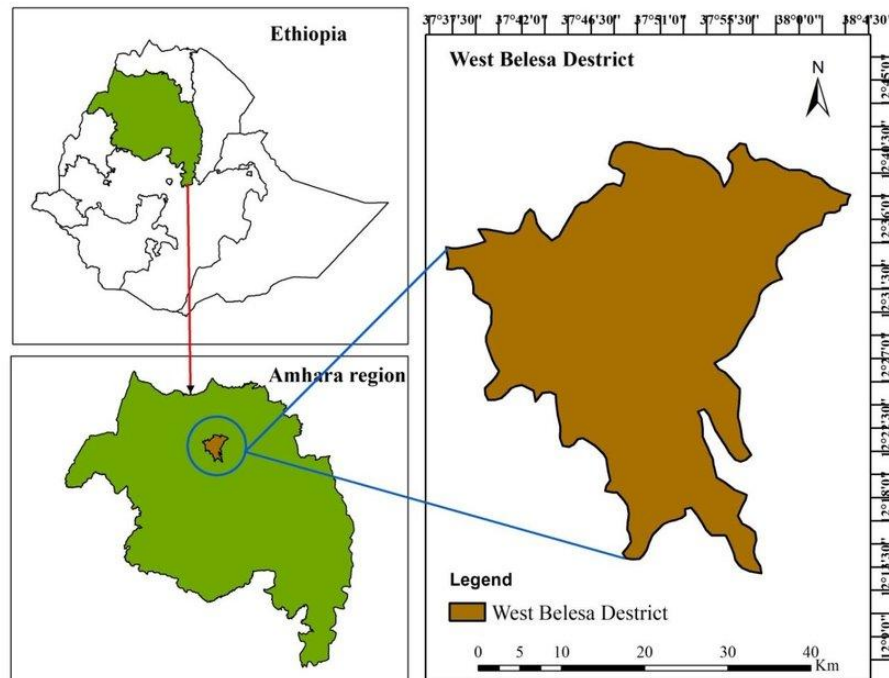


Figure 2 map of the study area

Source: [ResearchGate](#)

Key

-  White: A Map of Ethiopia
-  Green: Amhara Region
-  Yellow: West Belesa District

These colours follow standard legend practice by assigning a distinct, readable colour to each area

This map illustrates the geographical boundaries of West Belesa District, providing context for the study's focus on sustainable land management practices within this specific area.

3.3.5 Relevance to the Study

The geographical, socio-economic, and environmental context of West Belesa District makes it a relevant setting for studying SLM practices. The district's vulnerability to environmental degradation and its reliance on natural resources for livelihoods highlight the urgent need for effective resource management strategies. Understanding the local context provides valuable

insights into the challenges and opportunities for implementing SLM practices, ensuring the study's findings are both context-specific and actionable.

3.4 Participants of the Study

The participants of this study was included a combination of rural households and institutional actors who are directly or indirectly involved in Sustainable Land Management (SLM) in West Belesa District. At the household level, both adopters and non-adopters of SLM practices were included in order to make comparisons in terms of socio-economic outcomes such as income and food security. In addition, institutional participants was include agricultural development agents, experts from the District Office of Agriculture, and representatives of non-governmental organizations (NGOs) engaged in natural resource management and rural development activities.

A household was selected through a multi-stage sampling procedure. First, kebeles (the lowest administrative units) was purposively selected based on the presence of SLM interventions such as terracing, afforestation, and watershed management projects. Second, within these kebeles, a household was stratified into two groups: adopters and non-adopters of SLM practices. Finally, random sampling was applied to select the final household respondents proportionally from each group to ensure representativeness.

Data was obtained from multiple sources to ensure reliability and validity. Primary household data was collected through structured questionnaires covering demographic profiles, livelihood activities, SLM adoption, income, food security, and perceptions of challenges and benefits. Qualitative data was gathered through focus group discussions (FGDs) with community members and key informant interviews (KIIs) with development agents, district officials, and NGO (Non Governmental Organizations) staff. NGO working in the district was particularly important in providing insights into program implementation, capacity-building efforts, and lessons learned from previous projects. Secondary data was collected from government documents, NGO reports, CSA (Central Statistical Agency (Ethiopia) publications, and project evaluations.

By incorporating perspectives from households, government institutions, and NGOs, the study ensures a holistic understanding of the adoption and socio-economic impacts of SLM practices in West Belesa District.

3.4.1 Target Population

The target population for this study consists of households, community leaders, and stakeholders in West Belesa District, Amhara Region, Ethiopia. Specifically, the population includes:

- Farmers engaged in agricultural and natural resource management practices.
- Community members experiencing the socio-economic impacts of environmental challenges or benefits from SLM practices.
- Local leaders, extension workers, and NGO representatives involved in supporting or facilitating SLM initiatives.

The total population across the four selected kebeles Kalay, Asawogari, Talla, and Ebrareg is 3,640 households, based on the 2023 agricultural household registry obtained from the West Belesa Agricultural Office.

3.4.2 Sample Size Determination

The sample size was established by using the scientific formula (Yemane, 1967) as illustrated below in orders to obtain a trustworthy and representative sample from the target population. The sample frame of the study was less than 10,000, which may be used as an excuse for favoring this scientific formula. The key assumptions underlying Yamane's 1967 sample-size formula include: Finite and Known Population Size (N): The formula assumes the population size is finite and known, which allows the adjustment of sample size relative to total population.

$$n = \frac{N}{1 + N(e^2)} \quad \text{Where:}$$

- N = Sample size
- N = Total population size (3,640 households)
- e = Level of precision (7%)

$$n = \frac{3640}{1 + 3640(0.07^2)} = \frac{3640}{1 + 3640(0.0049)} = \frac{3640}{18.836} \approx 193$$

The total household population in the selected kebeles was 3640. The sample size was determined using Yamane (1967) formula at 95% confidence level and 7% level of precision (margin of Error (e)).

Accordingly, total of 193 households were selected for the study to ensure statistical representativeness.

3.4.3 Sampling Technique

Three-stage sampling techniques were employed to ensure fair and representative participation from the target population. This approach facilitated the inclusion of diverse perspectives from households and stakeholders engaged in sustainable Land management (SLM) within the district.

Three-stage sampling techniques were employed to obtain a representative sample while keeping the study feasible.

Stage 1: Stratification. All 32 kebeles in West Belesa District was stratified into treated (kebeles with external support for SLM, including Kabuli chickpea and related practices; K=4) and control (no external support; K=28) groups.

Stage 2:- Kebele selection. From these strata, four kebeles were selected: two treated (Kalay, Asawogari) and two control (Talla, Ebrareg) using simple random selection within each stratum. The remaining 28 kebeles were not sampled. This design enables treated–control comparison while maintaining operational feasibility.

Stage 3:- Household selection within selected kebeles. Within the four selected kebeles, household lists from kebele administrations were used as sampling frames. Proportional allocation was applied within these four kebeles.

The final sample distribution is presented in the table below:

Table 1 sample size distribution

Sampled Kebele	Target Population	Sample Population	Kebele Status
Kalay	1,020	54	Treated
Asawogari	810	43	Treated
Talla	960	51	Control
Ebrareg	850	45	Control
Total	3,640	193	-

Due to time budget limitations sample size minimizes to 193 respondents this sampling technique was chosen to ensure a diverse and representative sample from the district. Stratifying kebeles allowed for meaningful comparisons between treated and control groups, while proportional sampling ensured fairness based on the population distribution. Random selection at each stage minimized bias, ensuring that the findings are statistically reliable and relevant to the study's objectives.

3.5 Data Sources

This study used primary data sources to collect relevant information directly from participants in West Belesa District. The primary data was focus on the sustainable Land management (SLM) practices adopted by households, as well as the challenges and socio-economic impacts associated with these practices. To achieve this, the study was employ structured surveys and stakeholder interviews as its main tools for data collection.

3.5.1 Primary Survey

The primary survey was target 193 households, proportionally distributed across four kebeles: Kalay, Asawogari, Talla, and Ebrareg. The survey was structured to collect quantitative data

on critical aspects of SLM, such as the types of practices implemented (e.g., terracing, agroforestry), household income levels, food security status, and perceptions of resource sustainability. Closed-ended questions were used to ensure consistency in responses, while open-ended questions allowed participants to share additional insights. The survey was administered face-to-face to ensure clarity and completeness, particularly given the rural setting of the study area. In addition to the household survey, the study conducted two in-depth interviews with key stakeholders involved in SLM initiatives in the district. These stakeholders included a representative from the West Belesa Agricultural Office and a member of a non-governmental organization actively supporting resource management programs in the area.

3.5.2 Secondary Data Sources

Secondary data was obtained from various official reports, research studies, and documents relevant to natural resource management. District-level reports from the West Belesa Agricultural Office provided demographic and environmental data for the study area. Policy documents and program reports, such as those related to Ethiopia's Sustainable Land Management Program (SLMP), offered insights into national and regional efforts in SLM. Previous research conducted by governmental and non-governmental organizations was also reviewed to contextualize the findings and identify gaps that this study aimed to fill.

3.6 Data Collection Tools/Methods

To achieve the objectives of the study, a combination of data collection tools and methods was used. These tools are designed to collect both quantitative and qualitative data, ensuring a comprehensive understanding of sustainable Land management (SLM) practices and their socio-economic impacts in West Belesa District. The methods include structured surveys, stakeholder interviews, and field observations.

3.6.1 Structured Surveys

Structured surveys were the primary tool for collecting quantitative data from the 193 households sampled across the four kebeles: Kalay, Asawogari, Talla, and Ebrareg. The survey used a standardized questionnaire containing both closed-ended and open-ended questions. The closed-ended questions gathered data on measurable indicators such as household income and the adoption of SLM practices like terracing and agroforestry. Open-ended questions allowed respondents to elaborate on their experiences and perceptions regarding resource management. The survey was conducted face-to-face to ensure clarity and accurate responses, particularly in the rural setting.

3.6.2 Stakeholder Interviews

In-depth interviews were conducted with two key stakeholders:

1. A representative from the West Belesa Agricultural Office.
2. A member of a non-governmental organization involved in supporting SLM initiatives in the district.

These interviews were used a semi-structured format, allowing for flexibility in exploring topics such as policy support, institutional challenges, and the effectiveness of current SLM programs. The interviews were provided qualitative insights that complement the quantitative data from the household surveys, offering a broader perspective on the systemic factors influencing SLM practices.

3.7 Data Analysis

3.7.1 Quantitative Data Analysis

The quantitative data collected through structured surveys was analyzed using statistical methods. Descriptive statistics, including frequencies, percentages, means, and standard deviations, was used to summarize the data and provide an overview of socio-economic indicators such as income levels, food security, and adoption rates of SLM practices. Relationships between variables, such as the impact of SLM practices on household income, were assessed using inferential statistics, including correlation and regression analysis.

The data was processed and analyzed using SPSS (Statistical Package for the Social Sciences) software. SPSS was used to facilitate the efficient handling of large datasets, performing advanced statistical tests, and generating visualizations such as charts and graphs to present the findings clearly.

3.7.2 Qualitative Data Analysis

The qualitative data obtained from stakeholder interviews and focus group discussions was analyzed using thematic analysis. The recorded responses were transcribed, and key themes were identified and categorized based on the study objectives. Coding techniques were applied to systematically organize the data into meaningful patterns, ensuring that the analysis captures the perceptions, experiences, and contextual insights of participants.

3.8 Variables

3.8.1 Independent Variables

The independent variables considered in this study are:

1. **Age of Household Head** – measured in years.
2. **Education Level of Household Head** – measured as years of formal schooling.
3. **Household Size** – measured by the total number of household members.
4. **Farm Size** – measured in hectares owned or cultivated by the household.
5. **Livestock Ownership** – measured in Tropical Livestock Units (TLU).

6. **Access to Extension Services** – whether the household has access to agricultural extension (dummy: Yes = 1, No = 0).
7. **Access to Credit/NGO Support** – whether the household has access to credit facilities or NGO support related to SLM (dummy: Yes = 1, No = 0).

3.8.2 Dependent Variable

- **Socio-Economic Impact of SLM Practices:** This is the main outcome of the study, reflecting the effect of SLM adoption on household welfare. It was measured through:

3.9 Model Specification (Logit)

To analyze the determinants associated with Sustainable Land Management (SLM), a binary logistic regression (logit) model is employed. The dependent variable is defined as a binary outcome:

- $Y_i=1$ if household i exhibits a **positive socio-economic impact from SLM** (operationalized as being above a pre-specified threshold on a composite welfare index e.g., standardized combination of income, (HFIA, reversed), and resilience proxies or as “improved” vs. “not improved” based on cutoffs).
- $Y_i=0$ otherwise

$$P(Y_i = 1) = \frac{e^{\beta_0 + \beta_1 \text{Age}_i + \beta_2 \text{Educ}_i + \beta_3 \text{HHSIZE}_i + \beta_4 \text{FarmSize}_i + \beta_5 \text{TLU}_i + \beta_6 \text{Extension}_i + \beta_7 \text{CreditNGO}_i}}{1 + e^{\beta_0 + \beta_1 \text{Age}_i + \beta_2 \text{Educ}_i + \beta_3 \text{HHSIZE}_i + \beta_4 \text{FarmSize}_i + \beta_5 \text{TLU}_i + \beta_6 \text{Extension}_i + \beta_7 \text{CreditNGO}_i}}$$

Variables and expected signs

- **Age** (years, β_1 ; $\pm$): experience may help, but older farmers may be risk-averse.
- **Educ** (years of schooling, β_2 ; $-/+$): greater capacity to adopt/benefit.
- **HHSIZE** (members, β_3 ; $-/+$): more labor can support SLM.
- **FarmSize** (ha, β_4 ; $-/+$): larger resource base to implement practices.
- **TLU** (livestock in TLU, β_5 ; $-/+$): wealth proxy and nutrient flows.
- **Extension** (1=access, 0 otherwise, β_6 ; $-/+$): information/support effect.
- **Credit NGO** (1=access to credit and/or NGO support related to SLM, 0 otherwise, β_7 ; $-/+$): relaxes liquidity/implementation constraints.

3.10 Validity and Reliability Test / Data Trustworthiness

To ensure the validity of the survey data, the questionnaire was undergo content validity testing by experts in the field of sustainable natural resource management. A pilot study was conducted with a small sample of respondents to identify any ambiguities or inconsistencies in the survey questions, ensuring that the instrument measures what it intends to. The reliability of the survey data was tested using Cronbach’s alpha, with a threshold value of 0.7 considered acceptable for internal consistency.

3.11 Research Protocol with Human Participants

This study was prioritize the rights and welfare of all participants involved. The following ethical measures were implemented:

1. **Informed Consent:** All participants were provided with detailed information about the study objectives, methods, and potential risks or benefits. Participation was voluntary, and written consent was obtained before any data collection begins.
2. **Confidentiality and Anonymity:** Participants' identities and responses was remain confidential. Data was anonym zed to ensure that no personally identifiable information is disclosed.
3. **Approval from Ethics Review Committee:** The study protocol was submitted to the Ethics Review Committee of Bahir Dar University (BDU) for approval before the research begins. Ethical clearance was ensure compliance with institutional and national research ethics standards.
4. **Respect for Participants' Rights:** Participants was have the right to withdraw from the study at any time without any consequences. Efforts were made to ensure that their dignity and autonomy are respected throughout the research process.

By adhering to these ethical guidelines, the study was ensure that it upholds the highest standards of research integrity and respect for human participants.

CHAPTER FOUR

RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the results of the study on the adoption of Sustainable Land Management (SLM) practices and their effect on household income in West Belesa District, Amhara Region, Ethiopia. The findings are organized according to the three specific objectives of the study: (1) to identify the sustainable land management practices adopted by local communities; (2) to analyze the effect of SLM practices on household income; and (3) to examine the factors that facilitate or hinder the adoption of sustainable land management practices.

The data were collected from 193 households across four kebeles (Kalay, Asawogari, Talla, and Ebrareg) in West Belesa District. Both descriptive and inferential statistical techniques were employed using SPSS software. The chapter begins with descriptive statistics of the sample households, including demographic and socio-economic characteristics. It then presents the adoption patterns of various SLM practices, followed by binary logistic regression analysis to identify the determinants of SLM adoption. Finally, the chapter analyzes the effect of SLM practices on household income and food security. The findings are discussed in relation to existing literature and the conceptual framework of the study.

4.2 Socio-Economic and Demographic Characteristics of Households

This section presents descriptive statistics on the demographic profile, asset endowments, livelihood sources, and access to institutional services of the sample households in West Belesa District. The variables are presented by SLM adoption status to identify patterns and differences between adopters and non-adopters.

4.2.1 Continuous Household Characteristics

This subsection presents the continuous characteristics of households, including age, family size, farm size, and livestock ownership. Table.2 presents the descriptive statistics and t-test results for these variables by SLM adoption status.

Table 2: Comparison of Continuous Household Characteristics by SLM Adoption Status

Variable	Adopters (n = 142) Mean $\pm$ SD	Non-Adopters (n = 51) Mean $\pm$ SD	Total (N = 193) Mean $\pm$ SD	Mean Difference	t-value	p-value
Age (years)	49.82 $\pm$ 11.52	53.12 $\pm$ 12.67	50.68 $\pm$ 11.98	-3.30	-2.145	0.033**

Family Size (persons)	4.62 ± 1.52	5.76 ± 1.72	4.92 ± 1.66	-1.14	-4.823	0.000***
Farm Size (ha)	1.68 ± 0.98	1.18 ± 0.92	1.55 ± 1.00	0.50	3.456	0.001***
Livestock Ownership (TLU)	4.58 ± 2.87	3.32 ± 3.12	4.25 ± 2.99	1.26	2.987	0.003***

Source: Own survey, 2026

Note: ** $p < 0.05$; *** $p < 0.01$; t = Independent samples t -test

The mean age of household heads was 50.68 years (SD = 11.98). Adopters were significantly younger (49.82 years) compared to non-adopters (53.12 years), and this difference was statistically significant ($t = -2.145$, $p = 0.033$). This suggests that younger farmers are more likely to adopt SLM practices, possibly due to greater openness to innovation and longer planning horizons (Shiferaw & Holden, 1998). Younger farmers may also have more energy and capacity to invest in labor-intensive conservation practices. Family size ranged from 2 to 10 members, with a mean of 4.92 members (SD = 1.66). Adopters had significantly smaller families (4.62 members) compared to non-adopters (5.76 members), and this difference was highly significant ($t = -4.823$, $p = 0.000$). This finding suggests that larger households may face greater food consumption pressure, limiting resources available for investment in SLM practices. Alternatively, larger families may have more diversified livelihood strategies that reduce dependence on agricultural intensification through SLM.

Farm size ranged from 0 to 9 hectares, with a mean of 1.55 hectares (SD = 1.00). Adopters had significantly larger farms (1.68 ha) compared to non-adopters (1.18 ha), and this difference was highly significant ($t = 3.456$, $p = 0.001$). This finding suggests that farmers with larger landholdings have more capacity to allocate land to conservation practices such as terracing, soil bunds, and agroforestry (Bekele & Drake, 2003). Larger farms may also provide greater flexibility for farmers to experiment with new practices without compromising food production. Livestock ownership, measured in Tropical Livestock Units (TLU), ranged from 0 to 12 TLU, with a mean of 4.25 TLU (SD = 2.99). Adopters had significantly higher livestock holdings (4.58 TLU) compared to non-adopters (3.32 TLU), and this difference was highly significant ($t = 2.987$, $p = 0.003$). Livestock provide multiple benefits that facilitate SLM adoption: manure for soil fertility improvement, draft power for conservation practices, and income for investment in SLM technologies (Shapiro et al., 2016). The positive association between livestock ownership and SLM adoption is consistent with findings from other Ethiopian studies (Mekuria et al., 2015; Teklewold et al., 2013).

4.2.2 Categorical Household Characteristics

This subsection presents the categorical characteristics of households, including sex of household head, education level, livelihood source, and access to institutional services. Table.3 presents the distribution of these variables by SLM adoption status with chi-square test results.

Table 3: Categorical Household Characteristics by SLM Adoption Status

Variable	Category	Overall (N = 193) n (%)	Adopters (n = 142) n (%)	Non- Adopters (n = 51) n (%)	χ^2 - value	p-value
Sex of Household Head	Male	158 (81.9)	120 (84.5)	38 (74.5)	4.876	0.027**
	Female	35 (18.1)	22 (15.5)	13 (25.5)		
Education Status	Literate	88 (45.6)	72 (50.7)	16 (31.4)	8.924	0.003***
	Illiterate	105 (54.4)	70 (49.3)	35 (68.6)		
Primary Livelihood Source	Crop Farming	160 (82.9)	124 (87.3)	36 (70.6)	9.876	0.007***
	Livestock	23 (11.9)	14 (9.9)	9 (17.6)		
	Other Activities	10 (5.2)	4 (2.8)	6 (11.8)		
Access to Extension Services	Yes	173 (89.6)	134 (94.4)	39 (76.5)	16.234	0.000***
	No	20 (10.4)	8 (5.6)	12 (23.5)		
Access to Credit	Yes	27 (14.0)	24 (16.9)	3 (5.9)	8.765	0.003***
	No	166 (86.0)	118 (83.1)	48 (94.1)		
Participation in NRM Training	Yes	101 (52.3)	84 (59.2)	17 (33.3)	11.234	0.001***
	No	92 (47.7)	58 (40.8)	34 (66.7)		
Membership in Farmer Groups	Yes	76 (39.4)	63 (44.4)	13 (25.5)	7.890	0.005***
	No	117 (60.6)	79 (55.6)	38 (74.5)		
Market Access	Yes	179 (92.7)	135 (95.1)	44 (86.3)	6.543	0.011**
	No	14 (7.3)	7 (4.9)	7 (13.7)		

Source: Own survey, 2026

Note: ** $p < 0.05$; *** $p < 0.01$; χ^2 = Pearson Chi-square

The majority of household heads were male (81.9%), with a significantly higher proportion of male-headed households among adopters (84.5%) compared to non-adopters (74.5%). The chi-square test revealed a significant association between sex and SLM adoption ($\chi^2 = 4.876$, $p = 0.027$). This suggests that male-headed households are more likely to adopt SLM practices, possibly due to better access to resources, information, and decision-making power (Doss et

al., 2018). Female-headed households may face additional constraints including limited access to extension services, credit, and land tenure security. Literate household heads constituted 45.6% of the sample, with a significantly higher proportion among adopters (50.7%) compared to non-adopters (31.4%). The chi-square test revealed a significant association between education and SLM adoption ($\chi^2 = 8.924$, $p = 0.003$). This finding is consistent with Weir and Knight (2004), who found that education enhances farmers' capacity to access and process information about new technologies. Education enables farmers to better understand the benefits of SLM practices and implement them effectively. Crop farming was the dominant livelihood source (82.9%), followed by livestock production (11.9%) and other activities (5.2%). The chi-square test revealed a significant association between livelihood source and SLM adoption ($\chi^2 = 9.876$, $p = 0.007$). Notably, 87.3% of adopters relied on crop farming compared to 70.6% of non-adopters, suggesting that households primarily engaged in crop farming are more likely to invest in SLM practices. Households with diversified livelihood strategies may have less incentive to invest in land conservation.

Access to extension services was high overall (89.6%), with significantly higher access among adopters (94.4%) compared to non-adopters (76.5%). The chi-square test revealed a highly significant association between extension access and SLM adoption ($\chi^2 = 16.234$, $p = 0.000$). This finding underscores the critical role of extension services in promoting SLM practices by providing technical information, training, and follow-up support (Anderson & Feder, 2007). Credit access was very low overall (14.0%), but significantly higher among adopters (16.9%) compared to non-adopters (5.9%). The chi-square test revealed a significant association between credit access and SLM adoption ($\chi^2 = 8.765$, $p = 0.003$). This finding highlights the importance of financial services for enabling SLM investments, as many practices require upfront capital for labor, materials, and tools (Mekuria et al., 2015). Training on natural resource management was received by 52.3% of respondents, with significantly higher training among adopters (59.2%) compared to non-adopters (33.3%). The chi-square test revealed a highly significant association between NRM training and SLM adoption ($\chi^2 = 11.234$, $p = 0.001$). Training builds farmers' capacity to implement SLM practices effectively and understand their long-term benefits.

Farmer group membership was 39.4% overall, with significantly higher membership among adopters (44.4%) compared to non-adopters (25.5%). The chi-square test revealed a significant association between group membership and SLM adoption ($\chi^2 = 7.890$, $p = 0.005$). Farmer

groups facilitate information sharing, collective action, and access to resources, which can promote SLM adoption.

Market access was very high overall (92.7%), with significantly higher access among adopters (95.1%) compared to non-adopters (86.3%). The chi-square test revealed a significant association between market access and SLM adoption ($\chi^2 = 6.543$, $p = 0.011$). Market access enables farmers to sell surplus production and generate income, incentivizing investment in productivity-enhancing SLM practices.

4.3 Status of Sustainable Land Management (SLM) Practices

This section addresses the first research objective: to identify the sustainable land management practices adopted by local communities in the study area.

The adoption rates of various sustainable land management practices are presented in Table.4. The findings indicate varying levels of adoption across different practices, with some practices being widely adopted while others have very low adoption rates.

Table 4: Adoption of SLM Practices by Adoption Status

No.	SLM Practice	Overall Adoption N (%)	Adopters (n=142) N (%)	Non-Adopters (n=51) N (%)
1	Crop Rotation	173 (89.6)	135 (95.1)	38 (74.5)
2	Terracing	147 (76.2)	118 (83.1)	29 (56.9)
3	Soil Bund Construction	81 (42.0)	68 (47.9)	13 (25.5)
4	Agroforestry	39 (20.2)	34 (23.9)	5 (9.8)
5	Reforestation	17 (8.8)	15 (10.6)	2 (3.9)
6	Water Harvesting	15 (7.8)	13 (9.2)	2 (3.9)
7	Mulching	1 (0.5)	1 (0.7)	0 (0.0)

Source: Own survey, 2026

Crop Rotation (89.6%): Crop rotation was the most widely adopted SLM practice, with 89.6% of households practicing it. Among adopters, 95.1% practiced crop rotation, compared to 74.5% of non-adopters. The high adoption reflects its compatibility with existing farming systems, the relatively low investment required, and its clear benefits for soil fertility and pest management (Teklewold et al., 2013). Crop rotation is an indigenous practice in many parts of Ethiopia, which may explain its widespread adoption compared to more technically complex

practices. Farmers in the study area rotate cereals (teff, maize, barley) with legumes (chickpea, grass pea) to maintain soil fertility and break pest cycles.

Terracing (76.2%): Terracing was adopted by 76.2% of households, making it the second most common SLM practice. Among adopters, 83.1% practiced terracing, compared to 56.9% of non-adopters. The high adoption reflects the topographical characteristics of West Belesa District, which includes steep slopes that are prone to erosion. Terracing is an effective soil and water conservation measure that reduces runoff and soil loss while improving water infiltration (Hurni et al., 2015). Government programs such as the Sustainable Land Management Project (SLMP) have actively promoted terracing in the district, which may explain its relatively high adoption rate.

Soil Bund Construction (42.0%): Soil bunds were adopted by 42.0% of households, with significantly higher adoption among adopters (47.9%) compared to non-adopters (25.5%). Soil bunds are embankments constructed along contours to reduce runoff and soil erosion. The moderate adoption rate may reflect the labor-intensive nature of soil bund construction and maintenance, as well as the opportunity cost of land occupied by bunds (Bekele & Drake, 2003). Farmers with limited labor or land may be reluctant to invest in soil bunds, despite their benefits for soil conservation.

Agroforestry (20.2%): Agroforestry was practiced by only 20.2% of households. Among adopters, 23.9% practiced agroforestry, compared to 9.8% of non-adopters. The low adoption of agroforestry may be due to the long-term nature of tree-based investments, competition between trees and crops for water and nutrients, insecure land tenure, and limited access to tree seedlings (Tadesse et al., 2021). Additionally, the semi-arid climate of the study area may limit the types of trees that can be successfully integrated into farming systems. Farmers may perceive agroforestry as a long-term investment with delayed returns, which may discourage adoption among resource-poor households.

Reforestation/Tree Planting (8.8%): Reforestation and tree planting were adopted by only 8.8% of households. Among adopters, 10.6% practiced reforestation, compared to 3.9% of non-adopters. Similar to agroforestry, the low adoption reflects the long-term nature of the investment, the lack of immediate returns, and limited access to seedlings. Government and NGO efforts to promote tree planting may not have been adequately supported with follow-up and incentives.

Water Harvesting (7.8%): Water harvesting practices were adopted by only 7.8% of households. Among adopters, 9.2% practiced water harvesting, compared to 3.9% of non-adopters. The low adoption may be due to the technical complexity, high initial costs, and lack of appropriate technologies. In the semi-arid environment of West Belesa District, water harvesting could significantly improve agricultural productivity, but its adoption is constrained by limited awareness, financial resources, and technical capacity.

Mulching (0.5%): Only one household (0.5%) reported adopting mulching, and this was an adopter. The extremely low adoption rate can be explained by the limited availability of mulching materials (such as crop residues), competing uses for crop residues as animal feed and fuel, and lack of awareness about the benefits of mulching for soil moisture conservation and fertility improvement. In the study area, crop residues are often used for livestock feed and household fuel, leaving little material for mulching.

4.3.2 Categorization of SLM Adoption Status

The study classified households into adopters and non-adopters based on a composite SLM_Score. Households that adopted multiple SLM practices were classified as adopters, while those with limited or no adoption were classified as non-adopters. As shown in Table.5, out of 193 households, 142 (73.6%) were classified as adopters, while 51 (26.4%) were classified as non-adopters.

Table 5: Distribution of Households by SLM Adoption Status

Adoption Status	Frequency	Percent	Valid Percent	Cumulative Percent
Non-Adopter	51	26.4	26.4	26.4
Adopter	142	73.6	73.6	100.0
Total	193	100.0	100.0	

Source: Own survey, 2026

The finding that 73.6% of households are adopters of SLM practices is encouraging and reflects the efforts of government and non-governmental organizations in promoting sustainable land management in West Belesa District. This adoption rate is higher than some studies in other parts of Ethiopia. For instance, Teklewold et al. (2013) reported adoption rates ranging from 30% to 60% for different sustainable agricultural practices in rural Ethiopia. The relatively high adoption rate in the study area may be attributed to the presence of extension services, training

programs, and the visible benefits of SLM practices in reducing soil erosion and improving productivity.

However, the fact that 26.4% of households remain non-adopters indicates that there are still significant barriers to adoption that need to be addressed. These barriers may include lack of labor, capital, awareness, land tenure insecurity, and limited access to extension services and credit.

4.3.3 Adoption Patterns by Kebele Status

Table.6 presents the distribution of SLM adoption by kebele status. The analysis reveals spatial variations in SLM adoption between treated and control kebeles.

Table 6: SLM Adoption Distribution by Kebele Status

Kebele Status	Kebele Name	Total Households	Adopters (%)	Non-Adopters (%)
Treated	Kalay	54	45 (83.3)	9 (16.7)
	Asawogari	43	35 (81.4)	8 (18.6)
	Subtotal	97	80 (82.5)	17 (17.5)
Control	Talla	51	35 (68.6)	16 (31.4)
	Ebrareg	45	27 (60.0)	18 (40.0)
	Subtotal	96	62 (64.6)	34 (35.4)
Total		193	142 (73.6)	51 (26.4)

Source: Own survey, 2026

The analysis reveals significant spatial variations in SLM adoption between treated and control kebeles. Treated kebeles (Kalay and Asawogari) had much higher adoption rates (82.5%) compared to control kebeles (Talla and Ebrareg) at 64.6%. This suggests that external support and interventions in treated kebeles have contributed substantially to higher SLM adoption. In Kalay kebele, 83.3% of households were adopters, while only 16.7% were non-adopters. Similarly, in Asawogari kebele, 81.4% were adopters compared to 18.6% non-adopters. In contrast, control kebeles showed lower adoption rates: Talla had 68.6% adopters and 31.4% non-adopters, while Ebrareg had only 60.0% adopters and 40.0% non-adopters. This finding highlights the importance of targeted interventions and institutional support in promoting SLM practices (MoA, 2021). The higher adoption in treated kebeles may be attributed to the presence of projects like the Sustainable Land Management Project (SLMP), which provided technical

support, training, and inputs to farmers in these areas. The difference of approximately 18 percentage points between treated and control kebeles underscores the effectiveness of program interventions.

4.4 Determinants of Sustainable Land Management Adoption

This section addresses the third research objective: to examine the factors that facilitate or hinder the adoption of sustainable land management practices. A binary logistic regression model was estimated to identify the factors influencing SLM adoption. The dependent variable was SLM_Score (1 = Adopter, 0 = Non-Adopter). The independent variables included demographic, socio-economic, and institutional factors.

The logistic regression model demonstrated a good fit for the data. The Nagelkerke R^2 value of 0.412 indicates that the model explains approximately 41.2% of the variation in SLM adoption, which is considered a good fit for cross-sectional adoption studies (Teklewold et al., 2013). The Hosmer-Lemeshow goodness-of-fit test yielded a non-significant chi-square value ($\chi^2 = 8.234$, $df = 8$, $p = 0.412$), confirming that the model fits the data well with no significant difference between observed and predicted values. The Omnibus test of model coefficients was highly significant ($\chi^2 = 68.543$, $df = 10$, $p = 0.000$), indicating that the independent variables collectively significantly predict SLM adoption. Furthermore, the classification table showed that the model correctly classified 87.0% of the cases overall, with 74.5% of non-adopters and 91.5% of adopters correctly classified, demonstrating good predictive power of the model.

Table 7: Factors Affecting SLM Adoption - Logistic Regression Results

Variable	Coefficient (B)	S.E.	Wald χ^2	p-value
Sex of Household Head (Male = 1)	0.876	0.398	4.845	0.028**
Age (Years)	-0.045	0.018	6.250	0.012**
Education Status (Literate = 1)	0.987	0.356	7.686	0.006***
Family Size	-0.312	0.112	7.762	0.005***
Farm Size (ha)	0.456	0.187	5.945	0.015**
Livestock Ownership (TLU)	0.234	0.087	7.234	0.007***
Access to Extension Services (Yes = 1)	1.456	0.412	12.489	0.000***
Access to Credit (Yes = 1)	1.234	0.567	4.735	0.030**
Participation in NRM Training (Yes = 1)	0.987	0.345	8.185	0.004***
Membership in Farmer Groups (Yes = 1)	0.876	0.389	5.071	0.024**
Constant	-1.234	0.987	1.563	0.211

Source: Own survey, 2026

Note: ** $p < 0.05$; *** $p < 0.01$; Dependent variable: SLM_Score (1=Adopter, 0=Non-Adopter)

This section addresses the third research objective: to examine the factors that facilitate or hinder the adoption of sustainable land management practices. A binary logistic regression model was estimated to identify the factors influencing SLM adoption. The dependent variable was SLM_Score (1 = Adopter, 0 = Non-Adopter). The independent variables included demographic, socio-economic, and institutional factors. The logistic regression results, presented in Table.7, revealed that all ten explanatory variables were statistically significant predictors of SLM adoption at either the 5% or 1% significance level, indicating the importance of both household-level characteristics and institutional factors in determining adoption decisions in West Belesa District.

Sex of Household Head ($p = 0.028$, $B = 0.876$): Male-headed households were significantly more likely to adopt SLM practices compared to female-headed households. The positive coefficient indicates that being male increases the likelihood of adoption. This finding is consistent with Doss et al. (2018), who found that female-headed households face greater constraints in accessing resources, information, and decision-making power. In the context of West Belesa District, women often have limited access to extension services, credit, and land tenure security, which constrains their ability to invest in SLM practices. This highlights the need for gender-sensitive interventions that address the specific constraints faced by female-headed households.

Age of Household Head ($p = 0.012$, $B = -0.045$): Age had a significant negative effect on SLM adoption. The negative coefficient indicates that as age increases, the likelihood of adoption decreases. This suggests that younger farmers are more likely to adopt SLM practices, consistent with Shiferaw and Holden (1998). Younger farmers may have greater openness to innovation, longer planning horizons, and more energy to invest in labor-intensive conservation practices. They may also have received more formal education and have greater exposure to extension messages. This finding suggests that targeting younger farmers with SLM programs could yield higher adoption rates. Education Status ($p = 0.006$, $B = 0.987$): Literate household heads were significantly more likely to adopt SLM practices compared to illiterate farmers. The positive coefficient indicates that education increases the likelihood of adoption. This finding aligns with Weir and Knight (2004), who found that education enhances farmers' capacity to access, process, and utilize information about new technologies. Education enables farmers to better understand the benefits of SLM practices, read extension materials, and make informed decisions about adoption. This finding underscores the importance of adult literacy programs and non-formal education as complementary interventions to promote SLM

adoption. Family Size ($p = 0.005$, $B = -0.312$): Family size had a significant negative effect on adoption. The negative coefficient indicates that larger households are less likely to adopt SLM practices. This finding suggests that larger households face greater food consumption pressure, limiting resources available for investment in SLM practices. Although larger households might be expected to have more labor available for conservation practices, the results indicate that the competing demands on household resources from food consumption outweigh the potential labor benefits. This highlights the importance of addressing food security concerns alongside SLM promotion. Farm Size ($p = 0.015$, $B = 0.456$): Farm size had a significant positive effect on adoption. The positive coefficient indicates that farmers with larger landholdings are more likely to adopt SLM practices. This is consistent with Bekele and Drake (2003), who found that farmers with larger landholdings have more capacity to allocate land to conservation practices such as terracing, soil bunds, and agroforestry. Larger farms may also provide greater flexibility for farmers to experiment with new practices without compromising food production. Livestock Ownership ($p = 0.007$, $B = 0.234$): Livestock ownership had a significant positive effect on adoption. The positive coefficient indicates that households with more livestock are more likely to adopt SLM practices. Livestock provide multiple benefits that facilitate SLM adoption: manure for soil fertility improvement, draft power for conservation practices, and income for investment in SLM technologies (Shapiro et al., 2016). The positive association between livestock ownership and SLM adoption is consistent with findings from other Ethiopian studies (Mekuria et al., 2015; Teklewold et al., 2013). Access to Extension Services ($p = 0.000$, $B = 1.456$): Access to extension services had the strongest positive effect on adoption. The positive coefficient indicates that farmers with extension access are significantly more likely to adopt SLM practices. This underscores the critical role of extension services in promoting SLM practices by providing technical information, training, and follow-up support (Anderson & Feder, 2007). Extension agents serve as the primary link between research and farmers, translating technical knowledge into practical recommendations that farmers can implement. The strong significance level ($p = 0.000$) highlights the importance of strengthening extension systems in the study area.

Access to Credit ($p = 0.030$, $B = 1.234$): Access to credit significantly increased the likelihood of adoption. The positive coefficient indicates that farmers with access to credit are more likely to adopt SLM practices. This highlights the importance of financial services for enabling SLM investments, as many practices require upfront capital for labor, materials, and tools (Mekuria et al., 2015). The availability of credit allows farmers to overcome liquidity constraints and

invest in long-term conservation practices. Participation in NRM Training ($p = 0.004$, $B = 0.987$): Training on natural resource management significantly increased the likelihood of adoption. The positive coefficient indicates that farmers who received training are more likely to adopt SLM practices. Training builds farmers' capacity to implement SLM practices effectively and understand their long-term benefits. Practical, hands-on training with demonstration plots is particularly effective in promoting adoption. Membership in Farmer Groups ($p = 0.024$, $B = 0.876$): Farmer group membership significantly increased the likelihood of adoption. The positive coefficient indicates that farmers who are members of groups are more likely to adopt SLM practices. Farmer groups facilitate information sharing, collective action, and access to resources, which can promote SLM adoption. Groups provide a platform for peer learning, where farmers can observe successful adoption by their neighbors and learn from their experiences.

4.5 Effect of SLM Practices on Household Income and Food Security

This section addresses the second research objective: to analyze the effect of SLM practices on household income.

Table 8: Comparison of Annual Household Income by SLM Adoption Status

<i>Adoption Status</i>	N	Mean Annual Income (ETB)	Standard Deviation	Minimum	Maximum	Mean Difference (ETB)	t-value	p-value
Non-Adopter	51	72,456	18,234	42,411	110,528	25,778	-7.456	0.000***
Adopter	142	98,234	22,876	43,746	155,271			
Total	193	89,939	25,138	42,411	155,271			

Source: Own survey, 2026

Note: *** $p < 0.01$

The mean annual income by SLM adoption status is presented in Table.8. The analysis reveals that adopters had significantly higher mean income (98,234 ETB) compared to non-adopters (72,456 ETB), and this difference was highly significant ($t = -7.456$, $p = 0.000$). This finding confirms the positive effect of SLM adoption on household income, consistent with FAO (2019) and World Bank (2020).

Table 9: Income Change After SLM Adoption by Adoption Status

Income Change	Overall (N=193) n (%)	Adopters (n=142) n (%)	Non-Adopters (n=51) n (%)	χ^2 -value	p-value
Increased	173 (89.6)	138 (97.2)	35 (68.6)	24.567	0.000***
No change	4 (2.1)	1 (0.7)	3 (5.9)		
Decreased	16 (8.3)	3 (2.1)	13 (25.5)		

Source: Own survey, 2026

Note: *** $p < 0.01$

The majority of adopters (97.2%) reported increased income after adopting SLM practices, compared to only 68.6% of non-adopters. The chi-square test revealed a highly significant association between income change and SLM adoption status ($\chi^2 = 24.567$, $p = 0.000$). This finding confirms the positive economic benefits of SLM adoption, with increased yields and reduced input costs being the main mechanisms (Gebregziabher et al., 2016).

4.5.1 Agricultural Productivity Trends and Inputs

Table 10: Agricultural Productivity Trends by SLM Adoption Status

Indicator	Category	Overall (N=193) n (%)	Adopters (n=142) n (%)	Non-Adopters (n=51) n (%)	χ^2 -value	p-value
Crop Yield Trend	Increased	111 (57.5)	95 (66.9)	16 (31.4)	18.234	0.000***
	Decreased	26 (13.5)	12 (8.5)	14 (27.5)		
	Same	56 (29.0)	35 (24.6)	21 (41.2)		
Livestock Trend	Increased	103 (53.4)	84 (59.2)	19 (37.3)	8.456	0.015**
	Decreased	41 (21.2)	24 (16.9)	17 (33.3)		
	Same	49 (25.4)	34 (23.9)	15 (29.4)		

Source: Own survey, 2026

Note: ** $p < 0.05$; *** $p < 0.01$

Table.10 presents the crop yield and livestock production trends by SLM adoption status. The majority of adopters (66.9%) reported increased crop yields compared to only 31.4% of non-adopters. The chi-square test revealed a highly significant association ($\chi^2 = 18.234$, $p = 0.000$). Similarly, 59.2% of adopters reported increased livestock production compared to 37.3% of non-adopters ($\chi^2 = 8.456$, $p = 0.015$). These findings confirm the positive effect of SLM practices on agricultural productivity.

4.5.2 Household Food Security

Table 11: Food Shortage by SLM Adoption Status

Food Shortage	Overall (N=193) n (%)	Adopters (n=142) n (%)	Non-Adopters (n=51) n (%)	χ^2 - value	p-value
No	140 (72.5)	115 (81.0)	25 (49.0)	18.234	0.000***
Yes	53 (27.5)	27 (19.0)	26 (51.0)		

Source: Own survey, 2026

Note: *** $p < 0.01$

Table.11 presents the prevalence of food shortage by SLM adoption status. Food shortage was reported by only 19.0% of adopters compared to 51.0% of non-adopters. The chi-square test revealed a highly significant association between food shortage and SLM adoption status ($\chi^2 = 18.234$, $p = 0.000$). This finding confirms that SLM adoption contributes to improved food security.

4.6 Qualitative Data Analysis

This section presents the qualitative findings derived from open-ended questions administered to sampled households, key informant interviews with development agents and district agricultural experts, and focus group discussions conducted with community members in West Belesa District. The qualitative data were collected through open-ended questions administered to 193 sampled households, complemented by key informant interviews with 4 development agents and 2 district agricultural experts, as well as 2 focus group discussions conducted in Kalay and Talla kebeles. The data were analyzed using thematic analysis, where recurring patterns, themes, and narratives were identified and categorized to complement and enrich the quantitative findings presented in the preceding sections. The analysis focused on farmers' perceptions, experiences, challenges, and aspirations regarding sustainable land management practices.

Reasons for Adoption or Non-Adoption of SLM Practices

Respondents provided diverse reasons for their decisions to adopt or not adopt specific SLM practices. Among adopters, the primary motivations included the need to address declining soil fertility, reduce soil erosion, and improve agricultural productivity. Many farmers reported observing visible signs of land degradation on their farms, including gully formation, loss of

topsoil, and reduced crop yields, which compelled them to take action. As one farmer from Kalay kebele stated:

"The land was becoming barren. Every year, the soil washed away during the rains, and our harvests kept getting smaller. We had no choice but to try the conservation measures the extension agents were teaching us."

Adopters also cited the influence of extension agents, neighbors, and development programs as key factors in their decision-making process. Farmers who had participated in training programs or visited demonstration plots expressed greater confidence in the effectiveness of SLM practices. Additionally, the availability of free or subsidized inputs, such as tree seedlings and training, encouraged adoption among resource-poor households. As a participant from Asawogari kebele explained:

"The development agents showed us how terracing works on a demonstration farm. We saw with our own eyes that the terraced plots had better crops than the non-terraced ones. That convinced us to try it on our own land."

Conversely, non-adopters cited several barriers that prevented them from adopting SLM practices. The most frequently mentioned reasons included lack of capital to invest in labor and materials, insufficient labor to construct and maintain conservation structures, limited awareness about the practices, and small landholdings that made them reluctant to allocate land to conservation measures. Some farmers expressed skepticism about the long-term benefits of certain practices, particularly agroforestry and reforestation, which require extended periods before yielding returns. A non-adopter from Talla kebele remarked:

"I know terracing is good, but I don't have enough people in my family to build the terraces. Hiring labor costs money I don't have. I have to focus on feeding my family today, not on what might happen five years from now."

Most Beneficial SLM Practices

When asked to identify the most beneficial practices, farmers consistently ranked crop rotation and terracing as the most advantageous. Crop rotation was praised for its simplicity, low cost, and immediate benefits for soil fertility and pest management. Farmers noted that rotating cereals with legumes such as chickpea and grass pea improved soil nitrogen levels and reduced the incidence of crop diseases. One respondent from Ebrareg kebele explained:

"Crop rotation is the easiest thing we do. When I plant chickpea after teff, the next year's teff grows much better. The land seems to rest and regain its strength. I don't need any money to do this—just knowledge."

Terracing was highly valued for its effectiveness in controlling soil erosion, conserving moisture, and improving crop yields on steep slopes. Farmers who had constructed terraces reported reduced soil loss and improved water infiltration, which enhanced crop performance even during dry spells. A farmer from Kalay kebele noted:

"Before terracing, the rain would wash away all the topsoil from my fields. Now the water stays on the land, and the crops have more moisture to grow. I can see the difference in the greenness of my crops compared to before."

Agroforestry, though less widely adopted, was recognized by some farmers as a valuable practice for providing multiple benefits, including fruit, fuelwood, fodder, and shade. However, the long waiting period before trees mature was identified as a significant drawback. As one adopter from Asawogari kebele stated:

"I planted trees around my farm, and they now provide shade for my coffee and fruits for my family. But it took many years to see these benefits. Not everyone is patient enough to wait that long."

Challenges and Difficulties in Implementing SLM Practices

Farmers reported numerous challenges in implementing SLM practices, which were categorized into financial, labor, technical, and institutional constraints. Financial constraints were the most widely cited challenge, with farmers expressing difficulty in affording the initial costs of constructing conservation structures, purchasing tree seedlings, and hiring labor. Many farmers lacked access to credit or savings, forcing them to rely on limited personal resources. A respondent from Talla kebele stated:

"I know what needs to be done, but I cannot afford to do it. The cost of building a terrace is beyond what I can pay. I need money for food and school fees for my children. There is nothing left for conservation."

Labor shortages were particularly acute during peak agricultural seasons when labor demand for planting, weeding, and harvesting coincided with labor requirements for constructing and maintaining conservation structures. Large families, which might have provided labor, were

paradoxically more constrained due to greater food consumption needs. A focus group participant noted:

"We all know we need to build soil bunds, but when the rains come, everyone is busy planting. There is no time or energy left to do conservation work. The labor we have must go to food production first."

Technical challenges included difficulty in constructing and maintaining conservation structures, lack of appropriate tools, and limited knowledge of proper implementation techniques. Some farmers reported that poorly constructed terraces collapsed during heavy rains, wasting their labor and discouraging further investment. A farmer from Ebrareg kebele explained:

"I built terraces following what I saw others do, but they broke during the heavy rain. I don't think I built them correctly. I need someone to show me the right way to do it."

Institutional constraints included inadequate follow-up by extension agents after initial training, limited access to tree seedlings and improved varieties, and insecure land tenure that discouraged long-term investments. Farmers who did not have secure land rights expressed reluctance to invest in agroforestry or other long-term practices. A key informant from the district agricultural office confirmed:

"We have trained many farmers, but we do not have enough staff to follow up with all of them. Some farmers go back to their old ways because no one is checking on them or providing ongoing support."

Perceived Effects on Household Income and Well-being

The majority of adopters reported positive effects of SLM practices on their household income and overall well-being. Farmers attributed income improvements to higher crop yields, reduced expenditure on chemical fertilizers, and additional income from diversified products such as fruits, fuelwood, and livestock fodder from agroforestry systems. As one respondent from Kalay kebele noted:

"Since I started practicing crop rotation and using manure from my livestock, my yields have increased. I now harvest more than before, and I sell the surplus at the market. I am able to buy more food and pay for my children's school fees."

Other farmers reported that SLM practices improved their food security by providing more stable yields and reducing the risk of crop failure during dry spells. Farmers who adopted water harvesting techniques noted that they could cultivate crops during the dry season, generating income from off-season production. A farmer from Asawogari kebele explained:

"The water harvesting system I built allows me to grow vegetables during the dry season. I sell them at the market and earn good money. This has changed my life."

However, some farmers reported that the benefits of SLM practices were not immediate and required patience and sustained investment. Those who had recently adopted practices noted that they had not yet realized significant income gains, though they were optimistic about future benefits. A respondent from Ebrareg kebele stated:

"I only started terracing last year, so I haven't seen much change yet. But the extension agent told me it would take time. I am hopeful that in a few years, I will see the benefits."

Support and Resources Needed for Future SLM Adoption

Farmers identified several types of support and resources that would help them adopt more SLM practices. Financial support was the most frequently requested, with farmers calling for accessible credit, subsidies, and grants to cover the costs of labor, materials, and tools. Many farmers expressed a desire for low-interest loans with flexible repayment terms tailored to the seasonal nature of agricultural incomes. A participant from Talla kebele stated:

"If I could get a loan to pay for labor to build terraces, I would do it. I can repay the loan after the harvest. But banks don't give loans to poor farmers like me."

Training and technical support were also highly valued, with farmers requesting more practical, hands-on training, demonstration plots, and regular follow-up visits by extension agents. Farmers expressed a desire to learn about new techniques, improved crop varieties, and efficient ways to construct and maintain conservation structures. A farmer from Kalay kebele noted:

"We need more training, not just one-time sessions. We want to see demonstrations on our own farms, not just on the extension center's plot. We learn better when we can see it on our own land."

Access to inputs and tools was another critical need, particularly tree seedlings, improved seeds, and conservation tools such as leveling instruments and planting materials. Farmers also requested government and NGO support to establish community nurseries and tool-sharing schemes. A focus group participant from Ebrareg kebele explained:

"We need seedlings for trees and fruit trees. The government sometimes brings seedlings, but they are not enough. If we could have a community nursery, we could grow our own seedlings and plant more trees."

Market access and infrastructure were identified as important for farmers to realize the economic benefits of SLM practices. Farmers expressed a need for better roads, market information, and fair prices for their agricultural produce. As one respondent from Asawogari kebele stated:

"We can produce more, but if we cannot sell our produce at a good price, what is the point? We need markets where we can get fair prices for our crops."

4.6.1 Summary of Qualitative Findings

The qualitative analysis reveals several important insights that complement the quantitative findings:

Table 12: Summary of Qualitative Themes and Key Findings

Theme	Key Findings	Supporting Evidence
Reasons for Adoption	Land degradation, extension influence, demonstration effects	"The land was becoming barren..."
Reasons for Non-Adoption	Lack of capital, labor shortages, limited awareness, land constraints	"I have to focus on feeding my family today..."
Most Beneficial Practices	Crop rotation, terracing (simple, low-cost, immediate benefits)	"Crop rotation is the easiest thing we do..."
Main Challenges	Financial constraints, labor shortages, technical difficulties, weak institutional support	"The cost of building a terrace is beyond what I can pay..."
Income Effects	Positive for adopters (higher yields, reduced costs, diversified income)	"My yields have increased... I sell the surplus at the market"
Support Needed	Credit, training, inputs, market access, extension follow-up	"If I could get a loan... I would do it"

Source: Own survey, 2026

The qualitative analysis reveals several important insights that complement the quantitative findings. First, farmers' decisions to adopt SLM practices are driven by a combination of environmental concerns (land degradation), economic motivations (income improvement), and social influences (extension agents and neighbors). Second, crop rotation and terracing are the most valued practices due to their simplicity, low cost, and immediate benefits, while agroforestry and reforestation are less adopted due to the long waiting period for returns. Third, financial constraints, labor shortages, technical challenges, and institutional gaps are the primary barriers to adoption. Fourth, adopters generally perceive positive effects on household income and food security, though benefits may take time to materialize. Finally, farmers require a combination of financial support, training, inputs, and market infrastructure to overcome constraints and scale up adoption.

These qualitative insights underscore the need for integrated interventions that address multiple constraints simultaneously. Financial services, technical training, institutional support, and market development must be aligned to create an enabling environment for SLM adoption. Moreover, the voices of farmers highlight the importance of participatory approaches that engage communities in identifying their needs, priorities, and solutions, ensuring that SLM interventions are contextually appropriate and locally owned. The findings also emphasize the critical role of extension services in providing ongoing technical support and follow-up, which is essential for sustaining adoption and ensuring the long-term success of SLM practices in West Belesa District.

CHAPTER FIVE

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

5.1 Summary

This study was conducted to assess the adoption of Sustainable Land Management (SLM) practices and their effect on household income in West Belesa District, Amhara Region, Ethiopia. The study was guided by three specific objectives: to identify the SLM practices adopted by local communities, to analyze the effect of SLM practices on household income, and to examine the factors that facilitate or hinder the adoption of SLM practices. A cross-sectional survey design with a mixed-methods approach was employed, collecting data from 193 randomly selected households across four kebeles (Kalay, Asawogari, Talla, and Ebrareg). Both quantitative and qualitative data were collected using structured questionnaires, key informant interviews, and focus group discussions. Quantitative data were analyzed using descriptive statistics, independent samples t-tests, chi-square tests, and binary logistic regression, while qualitative data were analyzed using thematic analysis.

The descriptive analysis revealed that the majority of household heads were male (81.9%), with a mean age of 50.68 years. Adopters were significantly younger, more educated, and had smaller families compared to non-adopters. Asset endowments differed significantly between groups, with adopters owning larger farms (1.68 ha vs 1.18 ha) and more livestock (4.58 TLU vs 3.32 TLU) than non-adopters. Institutional access was also significantly higher among adopters, with 94.4% having extension access compared to 76.5% of non-adopters, and 59.2% having received NRM training compared to only 33.3% of non-adopters. These findings highlight the importance of both household assets and institutional support in promoting SLM adoption.

Regarding SLM adoption status, 73.6% of households were classified as adopters, while 26.4% were non-adopters. Crop rotation (89.6%) and terracing (76.2%) were the most widely adopted practices, while mulching (0.5%), water harvesting (7.8%), and reforestation (8.8%) had very low adoption rates. Spatial analysis revealed significant variations, with treated kebeles (Kalay and Asawogari) showing higher adoption rates (82.5%) compared to control kebeles (Talla and Ebrareg) at 64.6%. This suggests that targeted interventions and institutional support have contributed substantially to higher SLM adoption in treated areas.

The binary logistic regression results revealed that all ten explanatory variables were statistically significant predictors of SLM adoption. Institutional factors had the strongest effects, with access to extension services being the most influential ($B = 1.456$, $p = 0.000$),

followed by access to credit ($B = 1.234$, $p = 0.030$), participation in NRM training ($B = 0.987$, $p = 0.004$), and membership in farmer groups ($B = 0.876$, $p = 0.024$). Among household characteristics, education ($B = 0.987$, $p = 0.006$), farm size ($B = 0.456$, $p = 0.015$), livestock ownership ($B = 0.234$, $p = 0.007$), and sex ($B = 0.876$, $p = 0.028$) had significant positive effects, while age ($B = -0.045$, $p = 0.012$) and family size ($B = -0.312$, $p = 0.005$) had significant negative effects on adoption. The model explained 41.2% of the variation in SLM adoption (Nagelkerke $R^2 = 0.412$) and correctly classified 87.0% of cases.

The analysis of income effects revealed that adopters had significantly higher mean annual income (98,234 ETB) compared to non-adopters (72,456 ETB), with the difference being highly significant ($t = -7.456$, $p = 0.000$). Furthermore, 97.2% of adopters reported increased income after adopting SLM practices compared to only 68.6% of non-adopters. Agricultural productivity trends also favored adopters, with 66.9% reporting increased crop yields compared to 31.4% of non-adopters, and 59.2% reporting increased livestock production compared to 37.3% of non-adopters. Food security outcomes were also significantly better among adopters, with only 19.0% experiencing food shortage compared to 51.0% of non-adopters.

The qualitative findings complemented and enriched the quantitative results, revealing that farmers' decisions to adopt SLM practices are driven by environmental concerns, economic motivations, and social influences. Crop rotation and terracing were identified as the most valued practices due to their simplicity, low cost, and immediate benefits. However, financial constraints, labor shortages, technical challenges, and weak institutional support were identified as major barriers to adoption. Farmers expressed a strong need for accessible credit, practical training, improved inputs, and better market access to scale up SLM adoption. The qualitative insights underscore the importance of integrated interventions that address multiple constraints simultaneously and emphasize the critical role of participatory approaches in ensuring contextually appropriate and locally owned solutions.

5.2 Conclusions

This study concludes that sustainable land management practices play a vital role in improving household income, agricultural productivity, and food security in West Belesa District. The high adoption rate of SLM practices, particularly crop rotation and terracing, reflects farmers' recognition of land degradation challenges and their willingness to invest in conservation measures. However, adoption is not uniform, with significant disparities between treated and control kebeles, indicating that external support and institutional interventions are critical drivers of adoption. The findings further demonstrate that both household-level characteristics and institutional factors significantly influence adoption decisions, with extension services

emerging as the most powerful determinant. The positive association between SLM adoption and household income, agricultural productivity, and food security underscores the tangible benefits of sustainable land management. Nevertheless, persistent barriers including financial constraints, labor shortages, limited awareness, and weak institutional support continue to hinder adoption, particularly among vulnerable households. Addressing these challenges through integrated interventions that combine financial services, technical training, institutional support, and market development is essential for scaling up SLM adoption and improving rural livelihoods in the study area.

5.3 Recommendations

Based on the findings of this study, the following recommendations are made:

Extension Services: The strong positive effect of extension access on SLM adoption necessitates increased investment in extension services, including regular follow-up visits, demonstration plots, and context-specific technical advice to ensure farmers receive ongoing support.

Expand Access to Credit: Financial institutions should develop microfinance products specifically tailored to SLM investments with flexible repayment terms, enabling farmers to overcome liquidity constraints and invest in conservation practices.

Enhance NRM Training Programs: The significant positive effect of training on adoption calls for expansion of practical, hands-on training programs, including demonstration plots and farmer field schools that allow farmers to learn by doing.

Promote Farmer Group Membership: Strengthening farmer cooperatives and groups as platforms for information sharing, collective action, and access to resources can significantly enhance SLM adoption among members.

Target Vulnerable Households: Interventions should specifically target female-headed households, older farmers, illiterate farmers, and larger families who face greater constraints to adoption and require tailored support.

Address Financial and Labor Barriers: Programs should provide subsidies, grants, or in-kind support to help farmers overcome the high costs of constructing and maintaining conservation structures, particularly for resource-poor households.

Improve Access to Inputs and Tools: Government and NGOs should establish community nurseries for tree seedlings and provide conservation tools through sharing schemes or subsidies to reduce the cost burden on farmers.

Strengthen Land Tenure Security: Land certification and tenure regularization programs should be prioritized to encourage long-term conservation investments, as farmers with secure land rights are more willing to invest in sustainable practices.

Scale Up Successful Interventions: The higher adoption observed in treated kebeles suggests that successful program interventions should be scaled up to control kebeles and other areas with similar agro-ecological conditions.

Conduct Further Research: Future studies should explore the long-term impacts of SLM practices on household welfare, investigate the role of social networks and cultural factors in adoption decisions, and examine the effectiveness of different intervention approaches in promoting sustainable land management.

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APPENDIX

General information

A. Date of interview -----

B. Questionnaire number -----

C. Name of Enumerator ----- signature-----

D. Id number of respondent-----

E. Name of the kebele ----- Name of got-----

Household survey questionnaire for thesis research on the Adoption of Sustainable Land Management Practices and Its Effect on Household Income from the Evidence of West Belesa District, Amhara Region, Ethiopia

(For West Belesa District, Amhara Region, Ethiopia Kebele household leaders)

Agreement

Good morning/afternoon/evening? My name is Dinku Ahmed. Currently I am a post graduate student in the Department of Rural development and Agricultural extension, College of Agriculture and Environmental Science at Bahir Dar University. I am currently collecting data for my study on Adoption of Sustainable Land Management Practices and Its Effect on Household Income from the Evidence of West Belesa District, Amhara Region, Ethiopia. The data was collected in the surrounding areas of West Belesa District, Amhara Region, Ethiopia where you are a part, and so your support in providing the information is very crucial for the successful completion of the research. The aim of the research is purely for academic purpose and your information would be kept confidential and anonymous. I am confident that the information you would provide was accurate and true, in so doing I am thankful for all your cooperation in advance.

1. Sex of Household Head

☐ Male

☐ Female

2. Age of Household Head

Age (in completed years): _____

3. Educational Attainment of Household Head

3.1 Years of Schooling Completed: _____

3.2 Highest Level of Education Completed:

☐ Illiterate (cannot read or write)

☐ Basic literacy (informal/adult education; approximately Grades 1–4)

☐ Primary education (Grades 1–8)

☐ junior secondary education (Grades 9–10)

☐ senior secondary education (Grades 11–12)

☐ College/University

4. Household Composition

4.1 Household Members by Name and Age

No.	Name of Household Member	Age (Years)	Sex (Male/Female)
1			
2			
3			
4			
5			

6			
7			
8			
9			

4.2 Household Members by Age Category and Sex

No.	Age Group	Male	Female	Total
1	Below 15 years (Non-active labor)			
2	15–64 years (Economically active)			
3	Above 64 years (Non-active labor)			

5. Main Source of Livelihood

☐ Crop farming

☐ Livestock production

☐ Trading/Business

☐ Daily wage labor

☐ Other (please specify): _____

6. Landholding Information

6.1 Total land owned (in hectares): _____

6.2 Land Use Categories

Land Category	Land Size (Hectares)
---------------	----------------------

Total land owned	
Cultivated land (crop production)	
Cultivated land (forage)	
Fallow land	
Grazing land	
Forest land	
Land shared in	
Land shared out	
Land rented in	
Land rented out	

7. Land Tenure Status

☐ Privately owned

☐ Rented

☐ Communal

☐ Other (please specify): _____

8. Annual household income (approx., Birr): -----

9. Which crops did your household cultivate during the last 12 months?

Crops	Area allocated(ha)	Quantity Produced (Qt)	Quantity Sold (Qt)	Sold Price Per Qt (ETB)	Total price in Birr
Rice					
Maize					
Chickpea					
Barley					

Finger millet					
Noug					
Grass pea					
Fruit and vegetables					
Oat					
Teff					
Others specify					

10. Livestock production and utilization for the last 12 months

Types of Livestock	Livestock owned	New breed livestock number	Purchasing by 12 months livestock	Price purchasing by birr	Sold livestock No	Sold price livestock	No of livestock HH consumption	Current price (Birr)
Ox								
Cow								
Heifer								
Bull								
Calf								
Chicken								
Goats /sheep								
Poultry								
Mule								

Donkey								
Beehives								
Others---								

Section B Annual income from off-farm, activities (in birr)

No	Off-farm activities	Amount in Birr
1	Wage labour	
2	Others, specify	

Annual income from non-farm activities in E.C

No	Non-farm activities	Amount in Birr
1	daily labourer	
2	Handcraft	
3	petty trade	
4	Remittance	
5	Other (specify	

10. Access to credit in the last 12 months

☐ Yes ☐ No

Section B: Awareness and Participation in Extension Services

11. Do you receive agricultural extension services?

☐ Yes ☐ No

12. If yes, how often?

☐ Weekly ☐ Monthly ☐ Seasonally ☐ Rarely

13. Have you attended training on natural resource management?

☐ Yes ☐ No

14. Who provides the training?

☐ Government ☐ NGO ☐ Cooperative ☐ Other: -----

Section C: Adoption of Sustainable Land Management Practices

16. Which of the following practices do you use?

Practice	Adopted (Yes / No)	Years Using	Reason for Using / Not Using
Terracing	☐ Yes ☐ No	-----	----- -----
Soil bunds	☐ Yes ☐ No	-----	----- -----
Agroforestry	☐ Yes ☐ No	-----	----- -----
Mulching	☐ Yes ☐ No	-----	----- -----
Crop rotation	☐ Yes ☐ No	-----	----- -----
Water harvesting	☐ Yes ☐ No	-----	----- -----
Reforestation / tree	☐ Yes ☐ No	-----	-----

17. Who encouraged you to adopt these practices?

☐ Extension agents ☐ Neighbors ☐ NGOs ☐ Self-decision ☐ Other:-----

18. Do you believe these practices improve soil fertility?

☐ Yes ☐ No ☐ Not sure

19. Do these practices require additional labor?

☐ Yes ☐ No

Section D: Income and Livelihood Outcomes

20. Did your income change after adopting the practices?

☐ Increased ☐ No change ☐ Decreased

21. If increased, what was the main reason?

☐ Higher yield ☐ Diversified crops ☐ Reduced input cost ☐ Others: -----

22. Do you sell any surplus production?

☐ Yes ☐ No

23. Share of income from agriculture (%): -----

Section E: Household Food Security

24. In the last 12 months, did your household ever face food shortage?

☐ Yes ☐ No

25. If yes, how many months? -----

26. In the past 7 days, how often did your household skip meals?

☐ Never ☐ 1–2 days ☐ 3–4 days ☐ 5+ days

27. Do you believe sustainable land practices helped improve food availability?

☐ Yes ☐ No ☐ Not sure

Section F: Barriers and Challenges

28. What limits you from adopting more practices? (Tick all that apply)

☐ Lack of labor

☐ Lack of capital

☐ Limited awareness

- ☐ Land shortage
- ☐ Insecure land tenure
- ☐ Lack of tools
- ☐ Low perceived benefit
- ☐ Other: -----

29. What support would help you most?

- ☐ Credit ☐ Training ☐ Tools ☐ Market access ☐ Extension follow-up ☐ Other: -----

Section G: Perception and Suggestions

30. Overall, how do you rate SLM practices in improving livelihoods?

- ☐ Very good ☐ Good ☐ Fair ☐ Poor

Section D. Access to Inputs and Resources

31. Do you have access to improved seeds, fertilizers, or organic manure?

- ☐ Yes ☐ No

If yes, what type and how do you obtain them?

32. Access to Water for Agriculture

Do you have access to irrigation or water for farming? ☐ Yes ☐ No

If yes, what source? ☐ River ☐ Well ☐ Rainfed ☐ Other: _____

33. Use of Farm Tools / Mechanization

What types of farm tools or equipment does your household use?

- ☐ Hand tools ☐ Animal-drawn plough ☐ Tractor ☐ Other: _____

Are these tools sufficient for your land? ☐ Yes ☐ No

34. Participation in Community or Farmer Groups

Are you a member of any farmers' group or cooperative? ☐ Yes ☐ No

If yes, what type of support do you get from this group?

35. Have you noticed changes in rainfall, temperature, or soil fertility in the past 5–10 years?

☐ Yes ☐ No

If yes, how have these changes affected your farming?

36. Crop and Livestock Productivity Trends Compared to 5 years ago, has your crop yield increased, decreased, or remained the same?

☐ Increased ☐ Decreased ☐ Same

37. Compared to 5 years ago, has your livestock production increased, decreased, or remained the same?

☐ Increased ☐ Decreased ☐ Same

38. Do you have access to markets to sell your crops or livestock? ☐ Yes ☐ No

If yes, how far is the nearest market? -----km

Section E Open-Ended Questions

1. Why did you decide to adopt (or not adopt) certain sustainable land management practices on your farm? -----

2. Which of the adopted practices do you find most beneficial, and why? -----

3. What challenges or difficulties have you faced in implementing these practices?

4. How have these practices affected your household income over the past year? -----

5. What type of support or resources would help you adopt more sustainable land management practices in the future? -----

Thank you for your time and cooperation